

Membrane Construction Failure as a Unified Mechanistic Architecture in Autoimmune Disease:

From Sjögren's Syndrome to a Pan-Autoimmune Framework

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ABSTRACT

Background

Autoimmune diseases collectively affect more than 5% of the global population and represent more than 80 clinically distinct conditions, conventionally classified by target organ and effector immune mechanism. Despite divergent clinical phenotypes, an emerging body of mechanistic evidence suggests that shared upstream failures in barrier and membrane architecture precede and sustain autoimmune activation across this disease family. Current therapeutic paradigms, dominated by immunosuppressive and cytokine-blocking agents, address downstream immune effector stages while leaving primary barrier architecture defects unresolved, contributing to disease chronicity and incomplete remission.

Hypothesis

We propose a unified **Membrane Construction Failure (MCF)** framework in which the primary initiating event across diverse autoimmune diseases is the failure of tissue-specific physical barriers — epithelial, endothelial, synovial, nuclear, or myelin — to maintain proper compartmentalization. This failure leads to

aberrant immune antigen exposure, loss of self-tolerance, and the establishment of self-perpetuating inflammatory amplification loops that sustain disease independently of continued environmental provocation.

Methods

We conducted a systematic narrative synthesis of mechanistic literature spanning eight major autoimmune conditions: Sjögren's syndrome (SS), rheumatoid arthritis (RA), inflammatory bowel disease (IBD), multiple sclerosis (MS), psoriasis, celiac disease, systemic lupus erythematosus (SLE), and autoimmune thyroid disease (AITD). Primary literature was identified through PubMed, Embase, and Cochrane searches covering 2000–2025, supplemented by landmark pre-2000 mechanistic studies. Disease-specific barrier biology, molecular initiators, innate immune sensing mechanisms, and adaptive tolerance collapse pathways were extracted and mapped onto a proposed unified cascade architecture.

Results

Each of the eight diseases maps coherently onto a conserved five-stage MCF sequence: (1) Barrier assembly or maintenance failure; (2) Antigen spillover into immune-surveilled compartments; (3) Innate immune alarm activation; (4) Adaptive tolerance collapse; and (5) Autoantibody/autoreactive T-cell amplification loop with barrier feedback damage. Disease-specific molecular actors — aquaporin-5 (*AQP5*), tight junction proteins (claudins, occludin, ZO-1), zonulin, PAD4/citrullination machinery, blood-brain barrier occludin/claudin-5, flaggrin (*FLG*), and thyroid follicular cell MHC-II — are identified as tissue-specific executors of a common upstream failure mode. Shared genetic susceptibility loci (*HLA*, *PTPN22*, *IRF5*, *STAT4*, *IL23R*) are reframed as modulators of barrier integrity and antigen presentation thresholds rather than primary immune dysregulation genes. Pre-clinical barrier dysfunction is documented across all eight diseases, supporting MCF Stage 1 precedence over clinical autoimmunity.

Conclusions

The MCF framework reframes autoimmune disease as a family of barrier-engineering disorders with conserved upstream pathology and divergent downstream target-organ expression. Tissue tropism is determined by the

intersection of local barrier biology and HLA-restricted antigen presentation specificity, resolving the paradox of shared genetic risk producing phenotypically distinct diseases. This architecture has direct and underexplored therapeutic implications: barrier-restoration strategies acting upstream of immune activation constitute a largely neglected class of interventions with potential pan-autoimmune preventive efficacy. We propose a Barrier Integrity Index (BII) as a translational biomarker tool and outline a temporal combination strategy integrating barrier restoration with immune modulation.

Keywords: autoimmune disease; membrane construction failure; epithelial barrier; tight junction; aquaporin; citrullination; NETosis; blood-brain barrier; unified mechanism; barrier therapeutics; zonulin; type I interferon; PAD4; cGAS-STING; HLA restriction

1. Introduction

Autoimmune diseases collectively represent one of the most prevalent and clinically consequential categories of chronic illness in the industrialized world. More than 80 distinct conditions have been formally recognized under the autoimmune rubric, with a combined global prevalence exceeding 5% of the population [1,2]. By conservative estimate, this translates to approximately 400 million individuals worldwide living with some form of immunological self-attack, and epidemiological trends consistently indicate a rising incidence attributable to environmental exposures, altered microbiome composition, and epigenetic reprogramming rather than shifts in genetic population structure [3,4]. Incidence rates for rheumatoid arthritis (RA), inflammatory bowel disease (IBD), multiple sclerosis (MS), and autoimmune thyroid disease (AITD) have each increased measurably over the latter half of the twentieth century and into the twenty-first, with the steepest increases

occurring in industrialized nations undergoing the greatest ecological and dietary transformation [5].

The prevailing classification paradigm organizes autoimmune diseases principally by target organ — the joint in RA, the intestinal epithelium in IBD, the central nervous system in MS, the exocrine glands in Sjögren's syndrome (SS) — and by the predominant effector immune mechanism: autoantibody-mediated destruction, cytotoxic T-lymphocyte killing, immune complex deposition, or cytokine-driven tissue inflammation [6]. This organ-centric framework has proven enormously useful for the development of organ-specific diagnostics and has guided the rational design of biologics and small-molecule inhibitors targeting downstream effector pathways. Tumor necrosis factor-alpha (TNF- α) blockade in RA and IBD, anti-CD20 B-cell depletion in SLE and MS, and interleukin-17 (IL-17) neutralization in psoriasis and RA represent landmark therapeutic achievements of this paradigm [7,8].

Nevertheless, the organ-centric framework harbors a significant conceptual limitation: it is inherently downstream and reactive. The vast majority of approved autoimmune therapies target Stages 3–5 of disease pathophysiology — innate immune amplification, adaptive immune effector deployment, and autoantibody-mediated tissue damage — while leaving the upstream initiating events unaddressed [9,10]. The consequence is that discontinuation of immunosuppressive therapy reliably produces disease relapse, the burden of chronic immunosuppression accumulates in terms of infection risk, malignancy predisposition, and metabolic sequelae, and true immunological remission — the restoration of self-tolerance — is rarely achieved [11]. This therapeutic ceiling is not merely a technical limitation; it is, we argue, a consequence of not treating the disease at its origin.

The paradox that most forcefully demands a conceptual synthesis beyond organ-centric classification is the phenomenon of shared upstream architecture across phenotypically divergent diseases. Four lines of

epidemiological and genetic evidence converge on this point. First, familial co-aggregation studies consistently demonstrate that the first-degree relatives of patients with one autoimmune condition carry elevated risk not only for the same condition but for autoimmune disease in general [12]. Second, genome-wide association studies (GWAS) have catalogued a substantial repertoire of susceptibility loci shared across multiple autoimmune conditions, including the HLA class II locus (the single strongest genetic determinant of autoimmune risk), *PTPN22*, *IRF5*, *STAT4*, *IL23R*, and *CTLA-4* [13,14]. Third, patients with established autoimmune conditions carry substantially elevated rates of comorbid autoimmune diagnoses — a phenomenon known as autoimmune clustering that is far in excess of chance co-occurrence [15]. Fourth, therapeutic agents developed for one autoimmune condition demonstrably cross-class in efficacy: JAK inhibitors approved for RA work in IBD, psoriasis, and AITD; anti-IL-17 antibodies developed for psoriasis are effective in RA and ankylosing spondylitis; CTLA-4-Ig (abatacept) straddles RA, psoriatic arthritis, and AITD [16,17].

Collectively, these observations are irreconcilable with a model in which eight phenotypically distinct diseases represent eight fundamentally distinct pathological processes. They demand, instead, a model in which a common upstream vulnerability channels into organ-specific downstream expression.

The conceptual substrate for this synthesis has been building in the literature under several independent labels: the "leaky gut" hypothesis in IBD and celiac disease [18], the "barrier dysfunction hypothesis" in atopic dermatitis and psoriasis [19,20], the "blood-brain barrier failure model" in MS [21], and the "epithelial stress" model in SS [22]. What has been lacking is an integrative formal framework that unifies these disease-specific observations into a single mechanistic architecture applicable across the autoimmune spectrum and from which testable therapeutic predictions can be derived.

In this paper, we introduce and formally elaborate the **Membrane Construction Failure (MCF)** framework. We take as our prototype disease Sjögren's syndrome, in which the failure of salivary gland acinar cell membrane water-channel architecture — specifically aquaporin-5 (*AQP5*) downregulation driven by BMP6 overexpression — has been shown to initiate glandular dysfunction, antigen exposure, and lymphocytic autoimmune infiltration in a sequence that places barrier failure as the primary event, preceding and causing immune activation [22,23]. We then demonstrate that this disease-specific sequence is a tissue-specific instance of a universal five-stage MCF cascade operating across all eight diseases examined.

The objectives of this paper are: (1) to formalize the MCF framework with mechanistic precision; (2) to map the disease-specific molecular actors in each of eight autoimmune diseases onto the common MCF scaffold; (3) to synthesize the genetic and environmental determinants of tissue-specific barrier vulnerability; (4) to derive treatment implications from the framework's upstream logic; and (5) to propose a translational biomarker strategy — the Barrier Integrity Index — for clinical implementation of MCF-guided therapy.

We acknowledge at the outset the inherent limitations of a narrative synthesis approach. Cross-disease mechanistic comparison carries the risk of false analogical equivalence — a risk we attempt to mitigate by maintaining mechanistic specificity at each disease interface and by explicitly flagging where evidence quality is heterogeneous. The MCF framework is presented as a falsifiable mechanistic hypothesis generating specific testable predictions, not as a definitive settled account of autoimmune pathogenesis. We invite challenge, refinement, and experimental interrogation.

2. Conceptual Framework — The Membrane Construction Failure Model

2.1 Defining Membrane Construction Failure

We define **Membrane Construction Failure (MCF)** as any process — genetic, epigenetic, environmental, inflammatory, or metabolic — that impairs the assembly, maintenance, or dynamic repair of tissue-compartmentalizing barriers at the molecular, cellular, or structural level, resulting in loss of compartmental integrity sufficient to allow immune-relevant self-antigens to enter immune-surveilled space. This definition is deliberately broad with respect to etiology but mechanistically specific with respect to consequence: the critical output of an MCF event is antigen breach, not merely structural damage per se.

The term "membrane" in our framework encompasses both plasma membranes and supracellular barrier structures assembled from membrane proteins and their cytoskeletal scaffolds. We therefore include epithelial tight junction (TJ) networks, the myelin sheath, the nuclear envelope, the blood-brain barrier endothelial cell layer, and the cornified envelope of the epidermis within our definition of "membrane" — all share the property of being molecularly constructed barriers whose failure releases sequestered antigens into immune-accessible compartments.

We distinguish three classes of barrier in the human body that are mechanistically relevant to the autoimmune diseases under review:

Class A — Epithelial Barriers: These include the intestinal epithelium, the skin (stratum corneum and stratum granulosum tight junctions), the salivary and lacrimal gland acinar cell apical membranes, and the thyroid follicular epithelium. Epithelial barriers are constructed primarily through apical tight junction (TJ) complexes — transmembrane claudins, occludin, and junctional

adhesion molecules scaffolded by cytoplasmic ZO-1 and ZO-2 — and through specialized secretory or absorptive membrane architectures that serve barrier functions specific to their tissue context [24,25]. In salivary glands, the acinar cell apical membrane is also a secretory water-transport membrane whose integrity depends critically on aquaporin-5 (*AQP5*) channel density and localization [22].

Class B — Endovascular and Endothelial Barriers: These include the blood-brain barrier (BBB), the synovial microvasculature, and the glomerular filtration barrier. These barriers are constructed by highly specialized endothelial cells expressing brain- or tissue-specific TJ proteins (claudin-5, occludin at the BBB; claudin-1, PECAM-1 in synovial vessels) and pericyte-astrocyte support structures [21,26]. Their failure results in immune cell transmigration into immune-privileged compartments and in the deposition of immune complexes at critical filtration interfaces.

Class C — Intracellular Membrane Barriers: These include the nuclear envelope (constructed from lamins A/C, emerin, and nesprin network proteins), the mitochondrial outer membrane, and the myelin compaction interface. The nuclear envelope controls which nuclear antigens are exposed to cytoplasmic immune sensors (cGAS-STING, TLRs); the myelin sheath, assembled by oligodendrocytes through the compaction of myelin basic protein (MBP), proteolipid protein (PLP), myelin-associated glycoprotein (MAG), and myelin oligodendrocyte glycoprotein (MOG), acts as an electrical and immunological insulator whose disruption releases potent autoantigens [27,28].

2.2 The Five-Stage MCF Cascade

We propose that autoimmune disease initiation and perpetuation follows a universal five-stage cascade, with disease-specific molecular actors executing each stage within tissue-specific barrier contexts.

Stage 1 — Barrier Assembly or Maintenance Failure

The primary event is the loss of structural integrity at the tissue-specific barrier. This may be driven by: (a) genetic predisposition involving loss-of-function or hypomorphic variants in barrier structural genes (*FLG*, *CARD15/NOD2*, *AQP5* regulatory elements, claudin variants); (b) epigenetic silencing of barrier maintenance genes by environmental exposures; (c) a "second hit" — infection, dysbiosis, chemical irritants, hormonal perturbation, or nutritional deficiency — that overwhelms the reserve capacity of a genetically susceptible barrier; or (d) direct inflammatory damage to the barrier by locally produced cytokines that downregulate TJ gene expression via JAK-STAT, MLCK, or NF- κ B pathways [24,29,30]. Crucially, this stage may precede clinical autoimmunity by months to years, as documented in multiple disease contexts [18,31,32].

Stage 2 — Antigen Spillover and Compartmental Breach

Loss of barrier integrity allows normally sequestered self-antigens to enter immune-surveilled compartments for the first time, or at pathologically elevated concentrations that overwhelm peripheral tolerance mechanisms. These antigens include: nuclear DNA and chromatin (from ruptured nuclear envelopes in SLE); citrullinated self-proteins (from perforin-pored neutrophils in RA); myelin peptides (from complement-attacked myelin in MS); gliadin-TG2 complexes (from zonulin-opened intestinal TJs in celiac disease); bacterial LPS and flagellin (from TJ-breached intestinal epithelium in IBD); self-DNA/LL-37 complexes (from cracked epidermal barrier in psoriasis); thyroid peroxidase and thyroglobulin (from disrupted follicular tight junctions in AITD); and

AQP5/Ro/La antigens (from stressed salivary acinar cells in SS) [22,23,33,34,35,36,37,38].

Stage 3 — Innate Immune Alarm Activation

Released self-antigens act as damage-associated molecular patterns (DAMPs) detected by pattern recognition receptors (PRRs) of the innate immune system. The primary innate sensing pathways engaged include: TLR7 and TLR9 detecting single-stranded and CpG-rich double-stranded nucleic acids respectively; the cGAS-STING axis detecting cytoplasmic double-stranded DNA; the NLRP3 inflammasome detecting crystalline urate and mitochondrial DAMPs; TLR4 detecting LPS; and TLR2/TLR5 detecting bacterial structural patterns. Downstream consequences include NETosis (neutrophil extracellular trap formation) that amplifies DAMP release, complement pathway activation generating membrane attack complexes (MAC) that further damage barrier structures, NLRP3-driven IL-1 β and IL-18 production, and — critically — type I interferon (IFN- α/β) induction that acts as a systemic alarm signal maturing dendritic cells and lowering the threshold for adaptive immune activation [39,40,41].

Stage 4 — Adaptive Tolerance Collapse

Innate immune alarm signals mature dendritic cells (DCs) and upregulate MHC-II and co-stimulatory molecules (CD80, CD86, CD40), enabling presentation of previously ignored self-antigens to naïve CD4⁺ T cells in secondary lymphoid organs — now for the first time in an immunostimulatory rather than tolerogenic context. Autoreactive T cells

that escaped thymic clonal deletion (central tolerance) or peripheral regulatory mechanisms are activated, differentiated along Th1, Th17, or follicular helper T cell (Tfh) lineages, and licensed to provide B-cell help for autoantibody production. In parallel, regulatory T cell (Treg) function is suppressed by the pro-inflammatory cytokine milieu (TNF- α , IL-6, IL-1 β), and peripheral tolerance checkpoints are overwhelmed [6,42,43]. HLA class II polymorphisms determine which self-peptides are efficiently presented to autoreactive T cells — accounting for the HLA-disease associations that are disease-specific despite shared upstream barrier failure.

Stage 5 — Autoantibody/Autoreactive T-Cell Amplification Loop

The final stage establishes the self-perpetuating character of autoimmune disease. Disease-specific autoantibodies are produced by long-lived plasma cells and bind to their target antigens — on barrier structures themselves (anti-*AQP5*, anti-dsDNA, anti-tTG), on barrier-adjacent targets (anti-CCP, anti-MOG), or on functional membrane receptors (anti-TSH-R, anti-muscarinic receptor). These autoantibodies are not merely passive biomarkers; they are active agents of further barrier damage through complement fixation, Fc γ R-mediated effector activation, and direct receptor interference [44,45]. Autoantibody binding to barrier targets completes the feedback loop from Stage 5 back to Stage 1, producing the self-perpetuating nature of established autoimmune disease [9].

2.3 Why Barrier Failure Is Primary, Not Secondary

The most significant epistemological challenge to the MCF framework is the "chicken-and-egg" objection: is barrier dysfunction the cause of autoimmune

activation, or is it merely a secondary consequence of ongoing immune-mediated tissue inflammation damaging the barrier as a bystander? We address this objection through four independent lines of evidence that collectively establish Stage 1 barrier failure as temporally, genetically, and mechanistically upstream of adaptive immune activation.

Temporal precedence: Elevated intestinal permeability, measured by the lactulose-to-mannitol (L/M) ratio, is detectable in healthy first-degree relatives of IBD patients who have no clinical disease and no histological evidence of inflammation [32]. Similarly, subclinical BBB permeability changes, detectable by gadolinium-enhanced MRI, precede the formation of MS demyelinating plaques by weeks to months [46]. Pre-symptomatic salivary gland aquaporin redistribution and acinar stress markers appear in SS-risk individuals before lymphocytic focus scores become positive [23]. And the type I interferon signature that is the hallmark of SLE is detectable in peripheral blood up to 3–5 years before patients satisfy formal diagnostic criteria, but this IFN signature itself follows the appearance of anti-dsDNA antibodies that reflect nuclear antigen spillover — placing barrier failure (nuclear membrane clearance failure) upstream of the adaptive immune activation event [47].

Genetic evidence: Loss-of-function mutations in barrier structural genes are primary genetic risk factors for their respective autoimmune diseases, not epiphenomena of immune inflammation. Null mutations in the filaggrin gene (*FLG*) — encoding a structural protein essential for cornified envelope assembly in the stratum corneum — are the strongest known genetic risk factor for atopic dermatitis and are strongly associated with psoriasis and food allergy, conditions driven by epidermal antigen penetrance [48,49]. *NOD2/CARD15* mutations in Crohn's disease directly impair Paneth cell antimicrobial peptide secretion, driving dysbiosis and TJ stress — the *NOD2* gene product is a pattern recognition receptor expressed in epithelial barrier

cells, not in lymphocytes [50]. *DNASE1L3* and *C1Q* deficiency mutations in SLE directly impair nuclear membrane debris clearance, establishing the causal primacy of nuclear antigen spillover over adaptive immune activation [51].

Mechanistic dissociation in model systems: Perhaps the most compelling single piece of evidence for barrier failure primacy comes from the BMP6-transgenic mouse model of SS. Animals overexpressing BMP6 in salivary glands show suppressed *AQP5* transcription, reduced salivary flow, and acinar cell membrane dysfunction before any detectable lymphocytic infiltration or anti-Ro/La autoantibody generation. Subsequent gene therapy with adeno-associated virus delivering *AQP1* restores secretory function and simultaneously reduces the inflammatory infiltrate that develops downstream — demonstrating that the barrier failure drives the inflammation, not vice versa [22,23]. Analogously, in celiac disease, the administration of larazotide acetate (a competitive zonulin antagonist that prevents gluten-triggered TJ opening) reduces both intestinal permeability and gliadin-specific immune activation in human challenge studies — confirming that TJ opening is upstream of antigen-driven immune activation, not downstream [52].

Directionality from cytokine-TJ interaction: A common counter-argument holds that TJ proteins are downregulated by inflammatory cytokines (TNF- α , IL-1 β , IFN- γ) produced during immune activation, making TJ loss secondary to inflammation. While cytokine-driven TJ disruption is well-documented, this mechanism constitutes the Stage 5 $\rightarrow$ Stage 1 feedback amplification loop, not the initiating event. In the pre-disease state, TJ disruption occurs in the absence of significant local cytokine production, driven instead by primary barrier gene defects, dysbiosis-derived butyrate deficiency, or hormonal and nutritional perturbations that operate independently of adaptive immune activation [29,30,53].

3. Disease-Specific MCF Profiles

3.1 Sjögren's Syndrome — The Prototype MCF Disease

3.1.1 Primary Barrier Architecture

Sjögren's syndrome (SS) is a systemic autoimmune disease characterized clinically by exocrine gland dysfunction producing dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia), with systemic manifestations including fatigue, peripheral neuropathy, interstitial lung disease, and vasculitis in a significant minority of patients [54]. The disease affects an estimated 0.5–2% of the population with a striking female predominance (approximately 9:1 female-to-male ratio), and is the second most common systemic rheumatic disease after RA [54,55].

The **primary barrier** in SS is the salivary and lacrimal gland acinar cell apical membrane, and specifically the aquaporin-5 (*AQP5*) water channel network embedded within it. *AQP5* is a 26 kDa integral membrane protein belonging to the aquaporin family of transmembrane water channels and is expressed with high specificity in the apical plasma membrane of salivary and lacrimal acinar cells, where it mediates transcellular water flux essential for fluid secretion [56]. The structural integrity and appropriate membrane localization of *AQP5* is therefore not merely a secretory function; it constitutes a critical component of the acinar cell membrane architecture whose proper construction is required for normal glandular homeostasis.

3.1.2 MCF Initiator: BMP6-AQP5 Axis

The upstream MCF initiator in SS is the overexpression of bone morphogenetic protein 6 (BMP6) in the salivary gland microenvironment [22]. BMP6, acting through SMAD1/5/8 signaling, transcriptionally suppresses *AQP5* expression in acinar cells, leading to a reduction in apical membrane *AQP5* density and, critically, to the mislocalization of residual

AQP5 protein from the apical membrane to the cytoplasm — a form of membrane construction failure that impairs water transport even when total cellular *AQP5* protein levels are only moderately reduced [22,23]. This BMP6-driven *AQP5* suppression is detectable in salivary glands of SS patients and BMP6-transgenic mice before histological evidence of lymphocytic infiltration, establishing temporal primacy.

3.1.3 Downstream MCF Cascade in SS

Reduced transcellular water flux at the apical membrane leads to reduced mucosal surface hydration of oral and ocular epithelium. Mucosal desiccation induces epithelial stress responses including upregulation of heat shock proteins, translocation of normally nuclear SS-A (Ro) and SS-B (La) antigens to the cell surface during acinar cell apoptosis, and surface exposure of muscarinic M3 receptors in an altered conformational state [57,58]. These apoptosis-associated antigen exposures — particularly of the SS-A/Ro52 and SS-A/Ro60 ribonucleoprotein complexes — represent the Stage 2 antigen spillover event in SS.

Innate immune alarm activation (Stage 3) in SS is dominated by the type I interferon pathway: released SS-A/Ro ribonucleoprotein complexes containing endogenous RNA engage TLR7 on plasmacytoid dendritic cells (pDCs), triggering massive IFN- α production that matures conventional DCs and licenses Th17 and follicular helper T-cell differentiation [59]. The consequent adaptive immune response (Stage 4) generates the disease-defining autoantibodies: anti-Ro52 (TRIM21), anti-Ro60, and anti-La(SSB) IgG antibodies that are present in approximately 60–70% of primary SS patients [54].

Stage 5 autoantibody amplification in SS operates through two principal mechanisms: anti-muscarinic M3 receptor antibodies (anti-M3R) interfere with parasympathetic secretomotor signaling, compounding the *AQP5*-driven secretory failure with functional neurotransmitter blockade [58]; and anti-

AQP5 autoantibodies, detected in a subset of SS patients, directly promote AQP5 internalization from the apical membrane, further amplifying the Stage 1 barrier failure and completing the self-perpetuating damage loop [56].

3.1.4 Clinical Correlates and Extraglandular MCF

The clinical severity of sicca syndrome in SS patients correlates more closely with AQP5 membrane expression levels than with lymphocytic focus scores (the histological measure of immune infiltration) — providing direct clinical validation that the barrier (AQP5) failure, not the immune infiltrate, is the primary determinant of functional impairment [22,56]. Extraglandular SS manifestations — interstitial lung disease (reflecting alveolar epithelial barrier compromise), peripheral neuropathy (possibly reflecting vasculitic endothelial barrier disruption), and glomerulonephritis — suggest that the systemic type I IFN amplification initiated by salivary gland MCF propagates barrier-destabilizing inflammatory signaling to other tissue compartments. MCF classification: *Epithelial secretory membrane* → *water transport channel failure (AQP5 mislocalization)* → *acinar cell stress antigen release*.

3.2 Rheumatoid Arthritis — Synovial Membrane Perforin-Citrullination Failure

3.2.1 Primary Barrier Architecture

Rheumatoid arthritis is a chronic destructive autoimmune arthropathy affecting approximately 0.5–1% of the global population, characterized by symmetric synovitis, progressive joint destruction, and significant systemic comorbidity including accelerated cardiovascular disease [60]. The **primary barrier** in RA MCF is the synovial intimal lining membrane — a thin, non-vascularized layer of fibroblast-like synoviocytes (FLS) and macrophage-like synoviocytes (MLS) that normally forms a selective barrier between the joint cavity and the synovial vasculature, regulating articular fluid composition and excluding immune effector cells from direct cartilage contact. Additionally, the plasma membrane integrity of infiltrating neutrophils within

the pre-arthritic synovium constitutes a second barrier whose failure is mechanistically central to RA initiation.

3.2.2 MCF Initiator: Perforin-Mediated Plasma Membrane Poration and PAD4 Activation

The central MCF event in RA is the formation of poly-perforin pores in neutrophil plasma membranes by cytotoxic T lymphocytes (CTLs) activated within the synovial microenvironment [33,61]. Perforin, a pore-forming protein homologous to complement C9, oligomerizes in the target cell membrane to form transmembrane channels of 5–20 nm diameter. In the RA synovium, this CTL-to-neutrophil perforin attack constitutes a direct and dramatic plasma membrane construction failure — the pore itself being an absence of properly constructed membrane.

The intracellular consequence of plasma membrane poration is calcium influx, which activates peptidylarginine deiminase 4 (*PAD4*) — a calcium-dependent enzyme that converts arginine residues to citrulline in a post-translational modification (citrullination) that fundamentally alters the antigenic identity of intracellular proteins [33,62]. Mass citrullination occurs within perforin-pored neutrophils, targeting filaggrin, vimentin, fibrinogen, α -enolase, and histone H3 — proteins not represented in their citrullinated form during thymic T-cell education, making them neo-antigens capable of breaking tolerance [62,63].

3.2.3 Downstream Cascade: ACPA Generation and NETosis Amplification

Citrullinated peptides released from pored neutrophils are taken up by synovial dendritic cells and macrophages, processed through MHC-II (specifically HLA-DRB1 shared epitope alleles that have high-affinity binding pockets for citrullinated peptides), and presented to CD4+ T cells [13,63]. This generates anti-citrullinated protein antibodies (ACPAs) — the hallmark autoantibodies of RA — via Tfh-licensed B-cell activation. ACPAs form immune complexes with citrullinated synovial proteins, activating Fc γ RIIIa

on macrophages and complement on the synovial membrane, driving synovial inflammation and pannus formation (Stage 5).

A second and amplifying MCF pathway in RA is NETosis. IL-33 and immune complex stimulation activate RA neutrophils to undergo NETosis — a form of regulated cell death in which the nuclear envelope ruptures (an intracellular membrane failure), chromatin decondenses, and extracellular NET structures consisting of citrullinated histones, PAD4, elastase, and MPO are explosively released [40,64]. NETs are simultaneously a rich source of citrullinated autoantigens for ACPA generation and potent activators of TLR9 and complement — amplifying Stages 3 and 4 of the MCF cascade in a self-reinforcing manner.

3.2.4 Temporal Evidence: Pre-Clinical RA as MCF Stages 2-4

Among the most compelling evidence for MCF primacy in RA is the well-documented pre-clinical phase: ACPA positivity precedes the clinical onset of RA by a median of 5–7 years in prospective cohort studies, during which time patients have no clinical arthritis but display increasing ACPA titers, elevated CRP, and subclinical synovial thickening detectable on ultrasound [65,66]. This pre-clinical phase corresponds precisely to MCF Stages 2–4: antigen spillover (citrullinated peptide release), innate alarm (complement, TLR activation), and adaptive tolerance collapse (ACPA generation) — all occurring before Stage 5 clinical amplification becomes symptomatic. MCF classification: *Plasma membrane pore formation (perforin) → intracellular calcium → PAD4 activation → mass citrullination → neo-antigen release → ACPA generation.*

3.3 Inflammatory Bowel Disease — Intestinal Tight Junction Failure

3.3.1 Primary Barrier Architecture

Inflammatory bowel disease — encompassing Crohn's disease (CD) and ulcerative colitis (UC) — affects approximately 3 million individuals in North America and an estimated 6.8 million globally, with a rising incidence in previously low-prevalence regions tracking urbanization and dietary Westernization [67]. The **primary barrier** is the intestinal epithelial tight junction (TJ) complex — the most extensively studied paracellular seal in mammalian biology — constructed from claudin-1, claudin-2, claudin-3, claudin-4, claudin-7, occludin, and junctional adhesion molecules (JAMs) anchored to the cytoskeleton by ZO-1 and ZO-2 scaffolding proteins [24,25]. Overlying the TJ-sealed epithelium is the mucus bilayer — an inner dense layer produced by goblet cells that is essentially sterile and an outer loose layer colonized by commensal organisms — providing a first-line physicochemical barrier against luminal antigen access [68].

3.3.2 MCF Initiators: Parallel Genetic and Environmental Pathways

IBD represents a paradigmatic multi-initiator MCF disease in which genetic predisposition and environmental second hits operate in parallel pathways that converge on the same Stage 1 TJ failure outcome. The genetic pathway is best exemplified by *NOD2/CARD15* mutations, the strongest Crohn's disease genetic risk factor, which impair Paneth cell secretion of defensins and other antimicrobial peptides [50,69]. Reduced mucosal antimicrobial defense produces dysbiosis — specifically a loss of commensal bacteria that produce short-chain fatty acids (SCFAs, particularly butyrate) and an overgrowth of mucus-degrading and LPS-generating proteobacteria. SCFA deficiency directly reduces the expression of claudin-1, occludin, and ZO-1 in colonocytes via histone deacetylase (HDAC)-dependent mechanisms,

producing TJ failure as a downstream consequence of genetic barrier gene loss-of-function [29,53].

The environmental parallel pathway operates through dysbiosis-derived lipopolysaccharide (LPS), which engages TLR4 on the basolateral surface of enterocytes, activating myosin light chain kinase (MLCK) and causing perijunctional actomyosin contraction that physically forces TJ protein complexes apart — a mechanical form of membrane construction failure [29]. Additionally, loss-of-function variants in *PTPN2* (a protein tyrosine phosphatase that restrains STAT1 and STAT3 activation) lead to JAK-STAT hyperactivation and transcriptional downregulation of claudin-1 and ZO-1 — connecting the shared *PTPN2* risk locus directly to Stage 1 TJ failure [70].

3.3.3 Zonulin as the Molecular Gatekeeper

Zonulin — a peptide structurally related to complement factor B and zonula occludens toxin — has emerged as a critical endogenous regulator of intestinal paracellular permeability whose dysregulation constitutes a key MCF event in IBD [71]. Zonulin is released by intestinal epithelial cells in response to two principal stimuli: luminal gliadin peptides and dysbiotic gram-negative bacteria engaging CXCR3 on the epithelial apical surface. Zonulin acts via protease-activated receptor 2 (PAR2) signaling to trigger ZO-1 phosphorylation and disassembly from the TJ complex, creating transient but significant paracellular gaps through which luminal antigens — bacterial LPS, flagellin, food antigens — translocate into the lamina propria [71,72]. Serum zonulin levels correlate with both IBD disease activity and with claudin-1 protein expression levels in colonic biopsies, and predict histological disease severity — establishing zonulin as both a mechanistic intermediary and a potential Stage 1 biomarker [72].

3.3.4 Antigen Translocation and Innate-to-Adaptive Cascade

Translocated bacterial antigens encounter lamina propria macrophages bearing TLR4 (for LPS) and TLR5 (for flagellin), triggering NLRP3

inflammasome assembly, IL-1 β and IL-18 production, and TNF- α /IL-6 release [68,73]. This innate storm (Stage 3) drives DC maturation and Th1/Th17 polarization (Stage 4), generating the disease-specific autoantibody profiles that characterize CD (anti-Saccharomyces cerevisiae antibodies, ASCA; anti-OmpC antibodies) and UC (perinuclear anti-neutrophil cytoplasmic antibodies, p-ANCA) — Stage 5 adaptive outputs [6,73]. Critically, the pre-disease elevation of intestinal permeability in healthy first-degree relatives of IBD patients provides prospective evidence of Stage 1 precedence [32]. MCF classification: *Epithelial paracellular TJ seal failure (NOD2/dysbiosis/zonulin pathway) → luminal antigen translocation → NLRP3/TLR innate alarm → Th1/Th17 tolerance collapse.*

3.4 Multiple Sclerosis — Blood-Brain Barrier and Myelin Membrane Dual Failure

3.4.1 Primary Barrier Architecture

Multiple sclerosis is the most prevalent chronic neurological disease of young adults in high-income countries, affecting approximately 2.9 million people globally, and is pathologically characterized by focal demyelination of white matter tracts within the central nervous system [74]. The MCF framework in MS involves a dual barrier failure: **(A)** the blood-brain barrier (BBB) — a highly specialized endothelial barrier constructed from brain microvascular endothelial cells (BMECs) expressing the TJ proteins claudin-5, occludin, ZO-1, and ZO-2, supported by pericytes and astrocytic end-feet — and **(B)** the myelin sheath — a lipid-rich membrane structure assembled by oligodendrocytes through the compacted layering of MBP, PLP, MAG, and MOG around axonal segments [21,27,74].

3.4.2 BBB MCF Initiator: Systemic Inflammatory Destabilization of TJ Architecture

The initiating MCF event at the BBB is the destabilization of BMEC tight junction cytoskeletal anchoring by systemically derived inflammatory

cytokines — principally TNF- α , IL-17A, and IFN- γ — sourced from peripheral immune activation events including Epstein-Barr virus (EBV) reactivation, gut dysbiosis-driven systemic inflammation, or vitamin D deficiency-related impairment of VDR-mediated TJ gene expression [21,75,76]. These cytokines activate RhoA/ROCK and MLCK pathways in BMECs, disrupting actin microfilament attachment to the TJ scaffold and causing claudin-5 and occludin strand dissolution — measurably increasing BBB permeability to serum proteins and, critically, to immune cells [21,46].

EBV merits particular attention as an MCF trigger in MS: epidemiological and molecular evidence establishes EBV infection as virtually necessary for MS development, with EBV nuclear antigen 1 (EBNA-1) sharing structural homology with GlialCAM — an astrocytic adhesion molecule — enabling molecular mimicry-driven immune cross-reactivity that initially targets EBV-infected B cells harboring EBNA-1/GlialCAM homologs, and subsequently autoreactive CNS tissue [75,77]. This represents a pathogen-triggered TJ-destabilization mechanism feeding into MCF Stage 1 from an exogenous initiator.

3.4.3 Consequence of BBB Failure: Autoreactive T-Cell CNS Entry

A leaky BBB permits autoreactive Th1 (IFN- γ -producing) and Th17 (IL-17A-producing) cells — sensitized to MBP and MOG peptides in peripheral secondary lymphoid organs — to traverse the endothelial barrier into the CNS parenchyma [74,78]. CNS entry requires sequential molecular adhesion steps (LFA-1/ICAM-1, $\alpha 4\beta 1$ -integrin/VCAM-1) followed by endothelial diapedesis — steps that are gated by BBB TJ integrity. In the CNS, autoreactive T cells are reactivated by perivascular macrophages and activated microglia presenting myelin peptides on MHC-II, initiating the local immune attack on the myelin membrane [78,79].

3.4.4 Myelin Membrane MCF: Complement MAC and Oligodendrocyte Ferroptosis

The second MCF event in MS is direct attack on the myelin membrane. Complement proteins (C1q, C3, C5b-9 membrane attack complex) deposit on MOG on the outer myelin lamella surface — triggered by anti-MOG IgG antibodies — and form lipid bilayer pores (MAC) that destabilize the tightly compacted myelin membrane structure, causing unraveling of lamellae and exposure of interior myelin proteins including MBP and PLP [27,80]. This myelin membrane construction failure releases previously sequestered myelin peptides into the CNS extracellular space, where they drain via the glymphatic system and meningeal lymphatics to cervical lymph nodes — priming additional autoreactive T cells peripherally and establishing the outside-in/inside-out bidirectional amplification loop that characterizes progressive MS [80,81].

A further mechanism contributing to oligodendrocyte membrane failure is ferroptosis — a form of regulated cell death driven by iron accumulation and lipid peroxidation that specifically damages lipid bilayer integrity [82]. MS lesions show elevated iron deposition, and oligodendrocytes — the most metabolically iron-dependent cells in the CNS — are selectively vulnerable. Ferroptotic oligodendrocyte death releases intact myelin debris that overwhelms microglial phagocytic capacity, sustaining antigen availability and ongoing MAC-driven membrane attack. MCF classification: *Neurovascular endothelial TJ failure (BBB) → autoreactive T-cell CNS entry → complement MAC-mediated myelin membrane disruption → dual CNS barrier breach with bidirectional amplification.*

3.5 Psoriasis — Epidermal Barrier and Keratinocyte Membrane Failure

3.5.1 Primary Barrier Architecture

Psoriasis is a chronic, immune-mediated inflammatory skin disease affecting approximately 2–3% of the global population, characterized by hyperproliferative keratinocyte differentiation, parakeratosis, and plaques of erythematous, scaling skin [83]. The **primary barrier** is the stratified squamous epidermal barrier — comprising the cornified envelope of the stratum corneum (assembled from involucrin, loricrin, and filaggrin-processed natural moisturizing factors), the lamellar body lipid bilayer scaffold, and the stratum granulosum tight junctions (claudin-1, claudin-4, occludin) that constitute the so-called "second seal" of the epidermal barrier beneath the cornified layer [19,83,84].

3.5.2 MCF Initiator: Filaggrin Deficiency and Keratinocyte Differentiation Failure

Loss-of-function variants in the filaggrin gene (*FLG*) — present in approximately 10% of European populations — impair the post-translational processing of profilaggrin into filaggrin monomers that aggregate keratin intermediate filaments and are subsequently degraded into natural moisturizing factor (NMF) components [48,49,84]. *FLG* deficiency produces a structurally compromised stratum corneum with increased transepidermal water loss (TEWL) and physical microcracking — a form of cornified envelope construction failure. A parallel MCF pathway involves dysregulated keratinocyte differentiation — independent of *FLG* — in which the failure to properly assemble the cornified envelope leaves the stratum granulosum TJ network as the primary, unsupported barrier, which is more vulnerable to chemical and microbial disruption [83,84].

3.5.3 DAMP Signaling: LL-37 Self-DNA Complex and pDC Activation

Mechanically and osmotically stressed keratinocytes release a suite of alarmin signals including IL-33, thymic stromal lymphopoietin (TSLP), IL-1 α , and — critically — the cathelicidin antimicrobial peptide LL-37 [35,85]. LL-37 forms condensed complexes with self-DNA released from stressed keratinocytes — this represents a specifically barrier-failure-dependent DAMP: the self-DNA would not be released without keratinocyte membrane stress, and LL-37 would not form immunostimulatory complexes with it in the absence of barrier compromise that allows the two to co-localize in the extracellular space [85]. LL-37/self-DNA complexes penetrate endolysosomal membranes of plasmacytoid DCs and engage TLR7 and TLR9, triggering the type I IFN burst that matures conventional DCs and initiates Th17 commitment [85,86].

3.5.4 Th17/IL-23 Axis and Epidermal Barrier Self-Amplification

pDC-derived IFN- α activates conventional DCs to produce IL-23, which drives Th17 differentiation and IL-17A/F and IL-22 production by activated T cells in the psoriatic dermis [83,86]. Crucially, IL-17A and IL-17F complete the self-amplifying MCF loop by directly downregulating claudin-1 and occludin expression in keratinocytes, further compromising the stratum granulosum TJ seal and increasing epidermal permeability [87]. IL-22 drives keratinocyte hyperproliferation that prevents normal differentiation and cornified envelope assembly — mechanistically sustaining the Stage 1 barrier failure from a Stage 5 cytokine product. Single-cell RNA sequencing of psoriatic skin has further identified a subpopulation of keratinocytes undergoing partial epithelial-mesenchymal transition (EMT) that additionally lose E-cadherin and desmosomal adhesion proteins, expanding the paracellular permeability surface [88]. MCF classification: *Cornified envelope (FLG) and stratum granulosum TJ failure* $\rightarrow$ *LL-37/self-DNA DAMP complex* $\rightarrow$ *TLR7/9 pDC activation* $\rightarrow$ *type I IFN* $\rightarrow$ *IL-23/Th17* $\rightarrow$ *IL-17-mediated claudin suppression amplification loop*.

3.6 Celiac Disease — Gluten-Triggered Zonulin/Tight Junction Barrier Failure

3.6.1 Primary Barrier Architecture

Celiac disease is a chronic immune-mediated enteropathy triggered by dietary gluten (wheat gliadin and related proteins from rye and barley) in genetically susceptible individuals, with a global prevalence of approximately 1-2% [89]. It represents perhaps the most mechanistically tractable MCF disease, in that the environmental trigger (gluten), the barrier molecule (zonulin-regulated ZO-1/occludin TJ complex), the antigen (gliadin-TG2 complex), and the HLA restriction element (HLA-DQ2 or HLA-DQ8) are all precisely identified [52,71,89,90]. The **primary barrier** is the small intestinal epithelial tight junction complex, with particular emphasis on the zonulin-regulated opening mechanism.

3.6.2 MCF Trigger: Gliadin → Zonulin → TJ Disassembly

The MCF cascade begins when α -gliadin peptides — particularly the immunodominant 33-mer peptide resistant to complete gastrointestinal proteolysis — bind to CXCR3 on the apical surface of intestinal epithelial cells via a MyD88-dependent signaling pathway [71,90]. This engagement triggers the release of zonulin from epithelial cells into the intestinal lumen, where it acts in a paracrine manner on neighboring epithelial cells via PAR2 to phosphorylate ZO-1 and promote its dissociation from claudin and occludin anchoring complexes. The result is transient but significant paracellular gap widening — the opening of an otherwise intact barrier — through which intact and partially digested gliadin peptides translocate into the lamina propria [52,71,72].

3.6.3 Stage 2-4: HLA-DQ2/8 Restriction and tTG2 as Tolerance-Breaking Catalyst

Translocated gliadin peptides encounter subepithelial tissue transglutaminase-2 (tTG2) — an enzyme that deamidates glutamine residues

in gliadin peptides to glutamate, producing negatively charged peptides with approximately 100-fold increased binding affinity for HLA-DQ2 and HLA-DQ8 molecules on antigen-presenting cells [89,90]. This tTG2-mediated antigen modification is a critical Stage 2-to-4 amplifier: the native gliadin peptide is a suboptimal HLA-DQ2 binder, but the deamidated form binds with high affinity, enabling efficient CD4+ T-cell activation in the lamina propria. HLA-DQ2/8-restricted Th1-polarized CD4+ T cells produce IFN- γ , providing B-cell help for the production of anti-tTG2 IgA autoantibodies — the pathognomonic serological marker of celiac disease [89].

3.6.4 Autoantibody Amplification and Villous Destruction

Anti-tTG2 autoantibodies do not merely serve as diagnostic markers; they are active MCF amplifiers (Stage 5 $\rightarrow$ Stage 1 feedback). Anti-tTG2 IgA binds tTG2 on the surface of intestinal epithelial and endothelial cells, inhibiting TGF- β 1-dependent activation — a critical epithelial restitution signal — thereby impeding barrier repair independently of continued gluten exposure [44,90]. In parallel, IL-15 produced by stressed epithelial cells activates intraepithelial lymphocytes (IELs) expressing the activating receptor NKG2D; upregulation of NKG2D ligands (MICA, MICB) on stressed enterocytes triggers IEL-mediated epithelial killing, producing villous blunting and crypt hyperplasia [91]. The larazotide acetate (AT-1001) story provides definitive pharmacological validation: this competitive zonulin antagonist, by preventing gliadin-triggered TJ opening, reduces intestinal permeability, attenuates gliadin-specific immune activation, and improves symptom scores in Phase 2b trials — confirming that the TJ opening is mechanistically upstream of antigen-driven immune activation [52]. MCF classification: *CXCR3-mediated zonulin release $\rightarrow$ ZO-1/occludin TJ disassembly $\rightarrow$ gliadin translocation $\rightarrow$ tTG2 deamidation $\rightarrow$ HLA-DQ2/8-restricted tolerance collapse $\rightarrow$ anti-tTG IgA $\rightarrow$ impaired TGF- β 1 barrier repair $\rightarrow$ villous destruction.*

3.7 Systemic Lupus Erythematosus — Nuclear and Mitochondrial Membrane Failure → Type I IFN Storm

3.7.1 Primary Barrier Architecture

Systemic lupus erythematosus is a prototypic systemic autoimmune disease characterized by immune complex-mediated multi-organ involvement — most critically lupus nephritis — and by the production of an extraordinary diversity of autoantibodies directed against nuclear and cytoplasmic components [92,93]. With a global prevalence of approximately 40–50 per 100,000 and a female-to-male ratio of 9:1, SLE exemplifies the hormonal and genetic vulnerability of autoimmune barrier failure [92]. The **primary barrier** in SLE MCF is the nuclear envelope — specifically the structural integrity of the lamin A/C, emerin, and nesprin network that normally confines nuclear DNA, RNA, and chromatin within the nuclear compartment — and the mitochondrial outer membrane that sequesters mitochondrial DNA (mtDNA) from cytoplasmic immune sensors [39,51,93].

3.7.2 MCF Initiator: Failure of Apoptotic Nuclear Material Clearance

Under normal physiological conditions, apoptotic cells are rapidly engulfed by macrophages and dendritic cells through the process of efferocytosis, preventing the release of nuclear contents into the extracellular space. In SLE, this clearance mechanism is impaired by multiple converging defects: deficiency of DNase I (which normally degrades extracellular DNA), deficiency or dysfunction of complement C1q (which normally opsonizes apoptotic debris for phagocytosis), and failure of efferocytosis mediated by deficient MerTK and TIM-4 signaling [51,93,94]. The result is secondary necrosis of apoptotic cells — collapse of the nuclear envelope without clearance — releasing nuclear DNA, RNA, chromatin, histones, and ribonucleoprotein complexes into the extracellular milieu: the Stage 2 antigen spillover event in SLE.

3.7.3 NETosis as MCF Executor and cGAS-STING as Innate Amplifier

SLE is characterized by the presence of low-density granulocytes (LDGs) — a pathological neutrophil subset with an exceptionally high propensity for NETosis [40,95]. In SLE LDGs, nuclear membrane rupture occurs rapidly in response to immune complex and ROS stimulation, releasing chromatin, citrullinated histones H3 and H4, PAD4, LL-37, and — crucially — mitochondrial DNA (mtDNA) into extracellular NET structures [40,95]. The mtDNA component is particularly potent as an innate immune activator: oxidized mtDNA engages the cytoplasmic DNA sensor cyclic GMP-AMP synthase (cGAS), which synthesizes the second messenger 2'3'-cyclic GMP-AMP (cGAMP), activating STING (stimulator of interferon genes) → TBK1 → IRF3 phosphorylation → type I IFN (IFN- α/β) transcription [39,96].

TREX1 — a cytoplasmic DNA exonuclease that normally degrades ssDNA generated by reverse transcription and DNA replication intermediates — is mutated in a significant subset of familial SLE and in the monogenic interferonopathy Aicardi-Goutières syndrome [96,97]. Loss of TREX1 function allows cytoplasmic DNA to accumulate and constitutively activate cGAS-STING, directly confirming that nuclear membrane clearance failure (and its cytoplasmic DNA consequence) is the primary driver of SLE type I IFN activation, not a secondary phenomenon of immune inflammation [97].

3.7.4 IFN Storm, Autoantibody Diversification, and Vascular MCF Extension

The type I IFN produced by cGAS-STING/TLR7/TLR9-activated pDCs matures conventional DCs and licenses the broad autoantibody diversification characteristic of SLE — anti-dsDNA, anti-Smith, anti-Ro, anti-La, anti-histone, anti-phospholipid — reflecting the extraordinary diversity of nuclear compartment antigens released in the Stage 2 spillover event [92,93]. These autoantibodies form immune complexes that deposit in the glomerular basement membrane and small dermal vessels, activating complement and causing MAC-mediated endothelial barrier disruption — extending the MCF

principle to the vascular compartment and producing the cardinal end-organ complications of lupus nephritis and dermal vasculitis [44,93]. The IFN- α preclinical signature — detectable 3–5 years before SLE diagnosis — provides prospective temporal evidence placing the cGAS-STING/nuclear membrane MCF event upstream of clinical disease [47]. MCF classification: *Nuclear envelope clearance failure (DNase I / C1q / efferocytosis deficiency) + NETosis → cytoplasmic dsDNA/mtDNA → cGAS-STING → type I IFN storm → pan-autoantibody generation → immune complex-mediated vascular MCF.*

3.8 Autoimmune Thyroid Disease — Thyroid Follicular Cell Barrier and Aberrant MHC-II Antigen Display

3.8.1 Primary Barrier Architecture and Disease Phenotypes

Autoimmune thyroid disease (AITD) encompasses two principal phenotypes: Hashimoto's thyroiditis (HT), the most common cause of hypothyroidism in iodine-sufficient populations, and Graves' disease (GD), the most common cause of hyperthyroidism — together affecting an estimated 5% of the global population with a marked female predominance (7:1 female-to-male ratio) [38,98]. Despite producing opposite functional thyroid states, both conditions share the same upstream MCF event: failure of the thyroid follicular cell (TFC) barrier to maintain immune compartmentalization of thyroid antigens.

The **primary barrier** in AITD is the thyroid follicular architecture — a spherical structure composed of TFCs lining a central colloid-filled lumen. TFC apical tight junctions (claudin-3, claudin-4, occludin) normally prevent colloid leakage and confine thyroid antigen exposure to the basolateral surface under tolerogenic conditions [38,99]. Critically, TFCs under normal conditions do not express MHC-II molecules, and thyroid antigens (thyroid peroxidase, TPO; thyroglobulin, Tg; TSH receptor, TSH-R) are presented only by thyroidal dendritic cells in a tolerogenic context consistent with self-tolerance maintenance [99].

3.8.2 MCF Initiator: Oxidative Stress and Aberrant MHC-II Induction on TFCs

The MCF initiation event in AITD is the induction of aberrant MHC-II expression on TFCs themselves under conditions of oxidative stress and inflammatory cytokine exposure [38,99]. Environmental triggers — including iodine excess (which increases thyroid oxidative load via H₂O₂-dependent organification), selenium deficiency (reducing glutathione peroxidase-mediated protection against iodination-derived ROS), goitrogen exposure, and EBV or *Yersinia enterocolitica* infection (the latter sharing outer membrane protein OmpG homology with the TSH receptor, facilitating molecular mimicry) — converge with genetic predisposition (*HLA-DR3*, *HLA-DR4*, *CTLA-4* 49A/G polymorphism, *PTPN22* R620W variant) to produce IFN- γ -mediated upregulation of MHC-II on TFC surfaces [38,98,99].

This aberrant MHC-II expression transforms TFCs from passive barrier cells into active antigen-presenting cells — a fundamental MCF event in which the follicular cell not only fails to maintain compartmentalization but actively participates in presenting its own antigens to immune surveillance in an immunostimulatory context, entirely bypassing the tolerogenic DC-based antigen presentation that normally maintains thyroid self-tolerance.

3.8.3 HT vs. GD: Divergent Adaptive Responses to the Same MCF Event

In Hashimoto's thyroiditis, the adaptive response to TFC MHC-II-presented TPO and Tg peptides is Th1-skewed, driving cytotoxic CD8⁺ T-cell and NK cell infiltration and TFC destruction [98,99]. The progressive destruction of TFCs releases TPO and Tg into the circulation — anti-TPO antibodies (present in >95% of HT patients) and anti-Tg antibodies are generated, and these autoantibodies promote complement-mediated TFC apoptosis, amplifying the original MCF event in a Stage 5 → Stage 1 feedback loop that produces progressive hypothyroidism [98,100].

In Graves' disease, the adaptive response is Th2-skewed, with B-cell dominance generating anti-TSH-R stimulating antibodies (TRAbs) that constitutively activate the TSH receptor on TFCs, mimicking TSH and driving follicular hyperplasia and unregulated thyroid hormone production [98]. Spatial transcriptomic analysis of thyroid tissue in AITD has confirmed that TFC-stromal-lymphocyte crosstalk is spatially organized within follicular architecture, with CD74/MIF-CXCR4 axis signaling amplifying local immune co-stimulation [101].

3.8.4 Follicular Tight Junction Failure as Antigen Amplifier

Inflammatory disruption of follicular tight junctions (claudin-3, claudin-4) by infiltrating immune cells and their cytokines causes colloid leakage — effectively creating a second Stage 2 antigen spillover event that supplements the MHC-II-mediated antigen presentation route [99]. This follicular TJ disruption ensures that even as TFCs are being destroyed, their antigens continue to be released into the perifollicular immune microenvironment, sustaining the Stage 3–5 cascade independently of continued MCF-II induction. MCF classification: *TFC membrane oxidative stress → aberrant MHC-II induction → direct TFC antigen presentation + follicular TJ disruption → organ-specific tolerance collapse with either destructive (HT) or stimulatory (GD) autoantibody phenotype.*

4. Unified MCF Architecture — Cross-Disease Convergence

4.1 The Common Molecular Vocabulary of Barrier Failure

Having mapped each of the eight diseases onto the five-stage MCF cascade, we can now identify the common molecular vocabulary through which barrier failure is executed and amplified across phenotypically diverse autoimmune

conditions. This convergence is not metaphorical — it is precise at the level of specific protein families, enzymatic pathways, and signaling nodes.

Table 1 (described below) provides a cross-disease MCF profile summary that maps each condition onto its primary barrier type, molecular MCF initiator, principal DAMP released, innate sensing receptor engaged, disease-defining autoantibody, and proposed barrier biomarker. The seven-column structure of this table reveals striking inter-disease convergence: tight junction proteins appear as the primary barrier in five of eight diseases (IBD, celiac, MS-BBB, psoriasis, AITD); DAMPs comprising nucleic acid-protein complexes (self-DNA complexes, citrullinated histones, NET-associated RNA) appear across four diseases (SLE, RA, psoriasis, SS); and the type I IFN pathway is the converging innate alarm output across at least five diseases (SLE, SS, psoriasis, MS, RA-NETosis pathway).

The convergent roles of specific molecular actors are particularly striking. First, **tight junction proteins** (claudins, occludin, ZO-1) serve as primary or secondary MCF executors across all Class A and Class B barrier diseases. The same protein family — assembled by the same cellular machinery involving cytoskeletal anchoring, phosphorylation-regulated claudin strand assembly, and ZO-protein scaffolding — is the shared structural element whose failure unleashes autoimmunity across IBD, celiac, MS, psoriasis, AITD, and SS [24,25,29,52,87]. Second, the **PAD4/citrullination/NETosis** axis is mechanistically central to RA and SLE, and emerging evidence suggests NET-derived citrullinated antigens are also relevant in SS and psoriasis — potentially creating a convergent neo-antigen pool across multiple diseases in individuals with shared *PTPN22* and *IRF5* susceptibility alleles [33,40,64,95]. Third, **type I IFN** is the single most powerful cross-disease amplifier: it matures DCs, suppresses Treg function, enhances autoreactive B-cell survival, and promotes TJ-destabilizing cytokine production (IL-17, TNF- α) — acting as a systemic alarm that converts local MCF Stage 1 events into

systemic Stage 4 tolerance collapse across SS, SLE, psoriasis, MS, and increasingly RA [59,85,92,93,94,95]. Fourth, **HLA class II restriction** is the adaptive immune gate in celiac (HLA-DQ2/8), HT/GD (HLA-DR3/4), RA (HLA-DRB1 shared epitope), and MS (HLA-DR15) — the same class of molecules serving as the disease-specific filter that determines which self-antigens, once spilled from their barrier compartment, generate productive autoreactive T-cell responses [6,13,89,90].

Table 1. Cross-Disease MCF Profile Summary

Disease	Primary Barrier Type	MCF Initiator	Key DAMP Released	Innate Sensor	Disease Autoantibody	Proposed Barrier Biomarker
Sjögren's Syndrome	Salivary/lacrimal acinar apical membrane (AQP5)	BMP6 → AQP5 suppression/mislocalization	SS-A/Ro, SS-B/La ribonucleoprotein complexes	TLR7 (pDC)	Anti-Ro52, anti-Ro60, anti-La, anti-M3R	Serum anti-AQP5 Ab; salivary AQP5 by RT-PCR
Rheumatoid Arthritis	Synovocyte / neutrophil plasma membrane	CTL perforin pore → PAD4 activation → citrullination	Citrullinated peptides (vimentin, fibrinogen, histone H3), NETs	TLR9, FcγRIIIa, complement C1q	Anti-CCP (ACPA), anti-citrullinated vimentin, RF	Plasma citrullinated histone H3; serum PAD4 activity
Inflammatory Bowel Disease	Intestinal epithelial TJ (claudins, occludin, ZO-1)	NOD2 deficiency / dysbiosis / MLCK-LPS / zonulin	LPS, flagellin, bacterial DNA, food antigens	TLR4, TLR5, NLRP3	ASCA (CD), p-ANCA (UC), anti-OmpC	Serum zonulin; lactulose/mannitol ratio; fecal calprotectin

Disease	Primary Barrier Type	MCF Initiator	Key DAMP Released	Innate Sensor	Disease Autoantibody	Proposed Barrier Biomarker
Multiple Sclerosis	BBB endothelial TJ (claudin-5, occludin) + myelin membrane (MBP, MOG, PLP)	Cytokine-driven TJ dissolution + MAC-mediated myelin unraveling	MBP, MOG, PLP peptides; myelin lipids	Complement MAC, microglial TLR2/4	Anti-MOG IgG, anti-MBP; OCB in CSF	CSF neurofilament light (NfL); serum GFAP; gadolinium BBB enhancement
Psoriasis	Stratum corneum / stratum granulosum TJ (claudin-1, filaggrin)	FLG deficiency + keratinocyte differentiation failure → TEWL + microcracking	LL-37/self-DNA complexes; IL-33, TSLP	TLR7, TLR9 (pDC)	Anti-keratin IgG (variable); anti-LL-37 Ab	Transepidermal water loss (TEWL); stratum corneum claudin-1 expression
Celiac Disease	Small intestinal epithelial TJ (ZO-1, occludin; zonulin-gated)	Gliadin → CXCR3 → zonulin release → ZO-1 phosphorylation	Deamidated gliadin peptides, gliadin-tTG2 complexes	TLR2, NKG2D (IELs), NLRP3	Anti-tTG IgA, anti-DGP IgG, anti-gliadin IgA	Serum zonulin; L/M permeability ratio; anti-tTG IgA
SLE	Nuclear envelope (lamin network) + mitochondria	Efferocytosis failure / TREX1 deficiency →	dsDNA, mtDNA, chromatin, LL-37, ribonucleoproteins	cGAS-STING, TLR7, TLR9 (pDC)	Anti-dsDNA, anti-Smith, anti-Ro, anti-La,	Plasma cfDNA; IFN score; serum STING

Disease	Primary Barrier Type	MCF Initiator	Key DAMP Released	Innate Sensor	Disease Autoantibody	Proposed Barrier Biomarker
	Endothelial outer membrane	NETosis → nuclear/mtDNA spillover	ns		anti-phospholipid, anti-histone	activation markers
Autoimmune Thyroid Disease	TFC apical membrane + follicular TJ (claudin-3, claudin-4)	Oxidative stress → aberrant MHC-II induction on TFCs + follicular TJ disruption	TPO, thyroglobulin, TSH-R peptides	MHC-II (aberrant TFC presentation), complement	Anti-TPO IgG, anti-Tg IgG, anti-TSH-R (TRAb)	Anti-TPO titer; thyroid ultrasound; vascular reactivity; selenium status

Abbreviations: AQP5, aquaporin-5; ACPA, anti-citrullinated protein antibody; BBB, blood-brain barrier; cfDNA, cell-free DNA; CCP, cyclic citrullinated peptide; DGP, deamidated gliadin peptide; GFAP, glial fibrillary acidic protein; IEL, intraepithelial lymphocyte; L/M ratio, lactulose-to-mannitol ratio; MAC, membrane attack complex; MBP, myelin basic protein; MCF, membrane construction failure; MOG, myelin oligodendrocyte glycoprotein; mtDNA, mitochondrial DNA; NFL, neurofilament light chain; OCB, oligoclonal bands; pDC, plasmacytoid dendritic cell; PLP, proteolipid protein; TFC, thyroid follicular cell; TJ, tight junction; TPO, thyroid peroxidase; TRAb, TSH receptor antibody; tTG, tissue transglutaminase; TEWL, transepidermal water loss; ZO, zonula occludens.

4.2 Shared Genetic Architecture Supporting MCF

The observation that multiple autoimmune conditions share susceptibility loci has historically been interpreted through the lens of immune dysregulation — the shared loci are assumed to primarily regulate lymphocyte activation thresholds, cytokine production, or immune cell development. The MCF framework offers a reinterpretation: many of these shared loci regulate barrier integrity and antigen presentation thresholds simultaneously, and

their disease-modifying effect operates through the barrier layer as much as through lymphocyte biology.

PTPN22, encoding the lymphoid protein tyrosine phosphatase LYP, is associated with RA, SLE, AITD, IBD, and SS [13,14]. While classically characterized as a negative regulator of T-cell receptor signaling, *PTPN22* also modulates JAK-STAT signaling in epithelial cells — *PTPN2* (the epithelial isoform most closely related to *PTPN22*) restrains STAT1 and STAT3 phosphorylation in intestinal and thyroid epithelial cells, directly regulating claudin-1, occludin, and ZO-1 expression [70]. Loss-of-function variants in the *PTPN2/PTPN22* locus therefore act at the barrier level — upregulating STAT-driven cytokines that downregulate TJ proteins — not merely at the lymphocyte level [70,102].

IRF5 — the interferon regulatory factor 5 — is a risk locus for SLE, RA, SS, and IBD, and is a primary transcriptional regulator of type I IFN responses [103]. In the MCF framework, *IRF5* risk variants amplify the Stage 3 innate alarm (IFN production) following Stage 2 antigen spillover, lowering the threshold between barrier failure and adaptive tolerance collapse. *STAT4*, associated with RA, SLE, SS, and IBD, similarly amplifies downstream IFN- γ and IL-12 signaling from innate to adaptive transition [13,14,103]. *IL23R* variants, associated with IBD, psoriasis, and RA, regulate the IL-23/Th17 axis — whose inflammatory output (IL-17A/F) feeds back to directly downregulate epithelial TJ proteins, completing the Stage 5 $\rightarrow$ Stage 1 barrier damage feedback loop [87,102,104].

Collectively, the shared genetic architecture of autoimmune diseases preferentially clusters around three MCF-relevant functions: barrier protein expression regulation (*PTPN2/PTPN22*, *VDR*, *IL6ST*), antigen presentation specificity (HLA class II — providing tissue-specific disease direction within the common MCF framework), and innate alarm amplification threshold (*IRF5*, *STAT4*, *TREX1*, *DNASE1L3*). This is not the genetic architecture one

would predict from a primary lymphocyte dysregulation model — but it is precisely what the MCF framework predicts [13,14,51,103].

4.3 Environmental Modulators of MCF

Environmental factors that modify autoimmune risk consistently operate through the barrier layer, providing independent cross-disease validation of the MCF framework. **Microbiome dysbiosis** is associated with increased disease risk or activity in IBD, MS, celiac, SS, RA, psoriasis, and SLE, and in each case the mechanistic link operates through barrier integrity: dysbiotic microbiomes produce less SCFA (reducing TJ expression), more LPS and flagellin (activating MLCK-mediated TJ opening and TLR4 signaling), and less mucin-stimulating signaling (thinning the mucus barrier first line) [29,53,75,105]. **Vitamin D deficiency** increases risk for MS, RA, IBD, AITD, and SLE, and vitamin D receptor (VDR) signaling directly transcriptionally upregulates claudin-1, occludin, ZO-1, and filaggrin in multiple epithelial tissues — establishing VDR as a pan-MCF barrier maintenance transcription factor whose deficiency is a direct contributor to Stage 1 failure [76,106].

Hormonal factors deserve particular attention in explaining the striking female predominance of autoimmune diseases. Estrogen bidirectionally regulates multiple barrier systems: it upregulates claudin-5 expression in brain microvascular endothelial cells (protective of BBB integrity), downregulates intestinal claudin-1 during the luteal phase (increasing gut permeability cyclically), and modulates TFC MHC-II expression in thyroid tissue [107,108]. Estrogen also regulates *AQP5* expression in exocrine glands and modulates NETosis propensity in neutrophils — providing a hormonal MCF mechanism that plausibly contributes to female predominance across SS, SLE, AITD, MS, and IBD [107,108].

EBV infection is a cross-disease MCF trigger: it is associated with increased risk for MS (through EBNA-1/GlialCAM molecular mimicry and BBB TJ destabilization), SLE (EBV-driven type I IFN induction amplifying nuclear

antigen spillover responses), RA (EBV-encoded dUTPase activating TLR2 on synoviocytes), and SS (EBV reactivation in salivary gland epithelium driving TLR7-mediated IFN production) [75,77,109]. In each context, EBV operates as an environmental "second hit" that converts a genetically vulnerable barrier into an actively failing one.

4.4 The Amplification Loop: Why Autoimmunity Self-Perpetuates

A critical and distinctive feature of the MCF framework — compared with simple "loss of tolerance" models — is the explicit mechanistic account it provides for why autoimmune diseases are chronic and self-perpetuating. Once Stage 5 is reached, disease-specific autoantibodies are not merely diagnostic biomarkers of past immune activation; they are active participants in ongoing MCF maintenance through feedback destruction of barrier structures.

Anti-AQP5 antibodies in SS directly promote AQP5 internalization from the acinar cell apical membrane via receptor-mediated endocytosis — worsening the Stage 1 barrier failure that initiated the disease cascade and ensuring that removal of the original BMP6 stimulus would be insufficient to restore barrier function without autoantibody clearance [56,58]. ACPAs in RA activate complement on synovial endothelial cells, generating MAC-mediated endothelial barrier damage that releases further synoviocyte antigens and sustains the synovial inflammation driving continued PAD4 activation [44,62]. Anti-tTG2 IgA in celiac disease inhibits TGF- β 1-dependent epithelial restitution — impeding the repair of the very ZO-1/occludin TJ architecture whose disruption originally allowed gliadin translocation — meaning that the epithelium cannot heal even during gluten-free periods if anti-tTG2 titers remain elevated [44,90]. Anti-dsDNA immune complexes in SLE engage Fc γ RIIa and TLR9 simultaneously on pDCs in a synergistic co-receptor mode that drives sustained IFN- α production — the same IFN- α that matures DCs

and suppresses Treg function, ensuring that the Stage 3 innate alarm continues firing independently of new nuclear DAMP release [92,93,96]. Anti-TSH-R stimulating antibodies in GD produce constitutive TSH-R signaling, generating excessive intracellular ROS through NADPH oxidase activation, directly inducing TFC oxidative stress — the same environmental initiator that originally triggered aberrant MHC-II expression and began the AITD MCF cascade [98,99].

This Stage 5 → Stage 1 feedback is the mechanistic basis of autoimmune disease chronicity and is the principal reason why immunosuppressive therapies targeting Stages 3–5 alone produce disease control but rarely cure: the autoantibody-mediated barrier damage continues to feed Stage 1, and the barrier, untreated, continues to feed Stage 2 [9,10,44].

4.5 Tissue Tropism: Why Different Barriers Break in Different People

The MCF framework must address the key mechanistic question of tissue tropism: given the shared genetic and environmental vulnerability architecture identified above, what determines whether a given individual develops SS rather than MS, or RA rather than celiac disease? We propose a three-tier deterministic model of tissue tropism that is intrinsic to and predicted by the MCF framework.

Tier 1 — Tissue-specific barrier vulnerability determined by local microenvironment: Each barrier exists in a unique local milieu defined by pH, oxygen tension, microbial composition, humidity, mechanical stress, hormonal exposure, and blood flow dynamics. The intestinal epithelial TJ is exposed to the highest microbial antigen load in the body and is therefore maximally vulnerable to dysbiosis-driven MLCK activation [24,29]. The BBB is uniquely sensitive to circulating inflammatory cytokines because brain microvascular endothelial cells lack the perivascular adipose buffer that attenuates systemic inflammatory signaling in peripheral organs [21,46].

Salivary acinar AQP5 is under exquisite local transcriptional control via BMP6 signaling with minimal systemic regulatory input — making it disproportionately vulnerable to local gland microenvironmental changes [22,23].

Tier 2 — HLA allele specificity determines which spilled antigens are immunogenic: The universal MCF cascade generates Stage 2 antigen spillover in genetically vulnerable barriers, but not all spilled antigens generate productive adaptive immune responses. The antigen must bind with sufficient affinity to the individual's HLA class II variant for DC-T cell presentation. HLA-DQ2 and HLA-DQ8 have uniquely high affinity for deamidated gliadin peptides — making celiac disease effectively impossible without these alleles — while being poor binders for citrullinated peptides or myelin antigens [89,90,110]. HLA-DRB1 shared epitope alleles provide the peptide-binding groove geometry specifically suited for citrullinated peptides from vimentin and fibrinogen — directing RA rather than other TJ-failure phenotypes in individuals with synovial barrier vulnerability [13,63]. HLA-DR15 has high affinity for MBP and MOG peptides, restricting MS to individuals with this allele among those with BBB barrier vulnerability [110,111].

Tier 3 — Tissue-specific MCF gene expression: The barrier-specific molecular effectors of MCF are expressed in a highly tissue-restricted manner, ensuring that even systemic genetic variants produce only tissue-specific barrier failures. AQP5 is expressed almost exclusively in salivary and lacrimal acinar cells and alveolar type I cells — AQP5 downregulation by BMP6 produces SS, not IBD, because AQP5 is not expressed in the intestine [56]. Claudin-5 is expressed specifically by brain microvascular endothelial cells among all endothelial cell types — systemic claudin-5 genetic variants selectively compromise the BBB, not peripheral vascular barriers [21]. *FLG* is expressed exclusively in stratified squamous epithelium — *FLG* null

mutations produce epidermal barrier failure in skin, not in intestinal or thyroid epithelium [48,49].

The intersection of these three tiers — local microenvironmental vulnerability (Tier 1) × HLA-directed antigen presentation specificity (Tier 2) × tissue-restricted barrier gene expression (Tier 3) — produces the observed tissue tropism of each autoimmune disease as a deterministic output of a universal upstream MCF vulnerability, rather than as eight independent disease processes.

5. Treatment Implications of the MCF Framework

5.1 Paradigm Shift: Barrier Restoration as Primary Therapeutic Target

The therapeutic implications of the MCF framework represent its most significant translational contribution. Conventional autoimmune therapy — with remarkable pharmacological sophistication — targets Stages 3–5 of the MCF cascade. TNF- α blockade suppresses Stage 3 innate alarm amplification; anti-IL-17 antibodies interrupt Stage 5 cytokine-driven barrier damage; anti-CD20 B-cell depletion reduces Stage 5 autoantibody production; JAK inhibitors suppress the STAT-mediated cytokine signaling that constitutes the biochemical language of Stages 3–4 [7,8,16,17]. These are powerful and clinically validated interventions with transformative impact on patient outcomes. The MCF framework does not dismiss or diminish them.

However, the MCF framework makes an additional and currently underexplored prediction: that therapies targeting Stage 1 barrier restoration will accomplish what downstream immunosuppression cannot — interrupting the autoantibody-to-barrier-damage feedback loop (Stage 5 →

Stage 1) and potentially allowing immune tolerance to re-establish in tissue-specific niches where antigen exposure is reduced and regulatory T cells can re-expand in a lower-inflammation environment [9,10,52]. This prediction has been experimentally validated in the celiac disease model (larazotide acetate) and in the BMP6/SS mouse model (AQP1 gene therapy), and implies a broadly applicable therapeutic strategy whose implementation across the autoimmune spectrum remains largely unexplored [22,52].

5.2 Disease-Specific Barrier-Targeted Therapies

We review the current and emerging landscape of barrier-targeted interventions across the eight MCF diseases, distinguishing between those with established clinical evidence and those at the preclinical or early-trial stage.

Sjögren's Syndrome: The most mechanistically direct intervention in SS is restoration of AQP5-mediated secretory membrane function. Adeno-associated virus (AAV)-mediated delivery of *AQP1* to irradiation-damaged or BMP6-transgenic salivary glands in murine models restores salivary flow and simultaneously reduces lymphocytic infiltration — validating that barrier function restoration precedes and drives immune normalization [22,23]. Translational human trials are in development following Phase 1 safety assessment of AAV2-*AQP1* in patients with radiation-damaged salivary glands [112]. Anti-BMP6 monoclonal antibodies represent a pharmacological alternative to gene therapy for suppressing the upstream MCF initiator — clinical-stage compounds targeting related BMP pathways (e.g., luspaterecept, sotatercept) provide validated antibody scaffolds for BMP6 targeting. Anifrolumab (anti-IFNAR1), approved for SLE and under Phase 3 investigation in SS, targets the type I IFN amplification pathway at Stage 3, addressing the downstream consequence of AQP5 failure while leaving the upstream unaddressed [113].

Rheumatoid Arthritis: PAD4 inhibitors represent the most direct barrier-targeted intervention in RA, preventing the calcium/perforin-triggered citrullination cascade that generates neo-antigenic DAMPs [33,62]. GSK199 and Cl-amidine — preclinical PAD4 inhibitors — reduce ACPA generation and synovial inflammation in murine collagen-induced arthritis models and are progressing toward Phase 1 evaluation [114]. DNase I treatment and NET-disrupting strategies (neutrophil elastase inhibitors, PAD4 inhibitors in the NETosis context) directly address the extracellular NET-mediated antigen amplification pathway [40,64]. These approaches would be appropriately combined with existing DMARD or biologic therapy targeting downstream Stages 3–5 inflammation.

Inflammatory Bowel Disease: Larazotide acetate — the most clinically validated MCF-targeted therapeutic across all eight diseases — is a competitive zonulin antagonist that directly prevents dysbiosis- and antigen-triggered TJ opening [52,72]. Phase 2b trials in celiac disease have demonstrated its barrier-restoration mechanism in the intestinal epithelium; analogous application in IBD (where zonulin is similarly elevated and TJ failure similarly documented) constitutes a logical translational extension that is currently under investigation [72,115]. Fibroblast growth factor 21 (FGF21) analogs promote intestinal epithelial restitution and TJ protein expression recovery following inflammatory injury [115]. Fecal microbiota transplantation (FMT), by restoring SCFA-producing commensal populations, addresses the upstream dysbiosis driver of TJ failure and represents a microbiome-level Stage 1 intervention [68,105].

Multiple Sclerosis: Among approved MS therapies, natalizumab (anti-VLA-4/ $\alpha 4\beta 1$ -integrin) achieves its efficacy by preventing autoreactive T-cell transmigration across the leaky BBB — a mechanistic point that is precisely Stage 1-to-Stage 2 in the MS MCF cascade, and its exceptional clinical efficacy validates the centrality of BBB failure in MS pathogenesis [74,116].

S1P receptor modulators (siponimod, ozanimod) additionally stabilize BBB endothelial junctions through S1P signaling on brain endothelial cells, beyond their lymphocyte sequestration mechanism — representing genuine dual-mechanism barrier-immune therapy [116,117]. Remyelination strategies (opicinumab targeting LINGO-1, which inhibits oligodendrocyte differentiation and myelination) directly address the myelin membrane construction failure, attempting to re-build the myelin barrier whose disruption sustains the MS MCF cycle [117].

Psoriasis: The therapeutic success of IL-17A/F blockade (secukinumab, ixekizumab) in psoriasis is reframed by the MCF model as a Stage 5 → Stage 1 loop interruption: by blocking IL-17's claudin-1-downregulating effect on keratinocytes, these therapies partially restore stratum granulosum TJ integrity and reduce epidermal permeability alongside their direct anti-inflammatory effects [83,87]. Barrier-first topical strategies — ceramide-based physiologic lipid mixtures, humectants that restore stratum corneum hydration — address Stage 1 from a non-immunological direction and demonstrate utility in both disease management and prevention of atopic-to-psoriatic progression [84,118]. Tapinarof, an aryl hydrocarbon receptor (AhR) agonist approved for psoriasis, has been shown to restore barrier integrity by upregulating claudin-1, filaggrin, and loricrin expression — validating Stage 1 targeting as a therapeutic strategy with direct anti-psoriatic efficacy [118].

Celiac Disease: Beyond larazotide acetate, the developing therapeutic pipeline for celiac disease includes several MCF-targeted approaches [52,89]. Endopeptidase-based gluten detoxification (ALV003 — a combination of a glutenase and prolyl endopeptidase; IMGX003 — an oral glutenase) degrades the 33-mer gliadin peptide before it can engage CXCR3 and trigger zonulin release — a Stage 0 intervention (before Stage 1 MCF occurs) that prevents the cascade entirely [119]. IMU-856 (a SIRT6 activator) promotes

intestinal epithelial regeneration and barrier restoration, addressing the failure of anti-tTG2-inhibited epithelial restitution [120]. The conceptual success of the celiac therapeutic pipeline — organized around blocking gluten-triggered TJ opening and restoring epithelial barrier integrity — provides the most compelling proof-of-concept for pan-MCF barrier-targeted therapy across all eight diseases.

SLE: TREX1 restoration — either by TREX1 enzyme replacement or by gene therapy — would address the primary nuclear membrane clearance failure in familial SLE, preventing cytoplasmic DNA accumulation and cGAS-STING activation [96,97]. While not yet in clinical development, STING antagonists (MSA-2, H-151) targeting the cGAS-STING pathway at Stage 3 represent a barrier-failure-specific innate suppressor that differs mechanistically from conventional immunosuppression [96,121]. Belimumab (anti-BAFF), approved for SLE, reduces Stage 5 autoantibody production by eliminating survival signals for autoreactive B cells; in the MCF framework, this is specifically the Stage 5 → Stage 1 loop interruption mechanism [93,121]. DNase I replacement/augmentation strategies — addressing the primary efferocytosis failure that allows nuclear material spillover — are under investigation in SLE models and represent the most direct Stage 1 intervention in this disease context [94].

Autoimmune Thyroid Disease: Selenium supplementation — at 200 µg/day — significantly reduces anti-TPO antibody titers in HT patients in randomized controlled trials, likely through restoration of glutathione peroxidase-mediated protection against TFC oxidative stress (the MCF Stage 1 initiator in AITD) [38,100,122]. This is currently one of the highest evidence-grade Stage 1 MCF-targeted interventions across all eight diseases. CTLA-4-Ig (abatacept) blocks CD80/CD86 co-stimulation required for MHC-II-restricted T-cell activation at Stage 4, interrupting the aberrant TFC antigen presentation cascade [16]. Retinoid receptor agonists (RAR/RXR ligands)

have been shown to downregulate IFN- γ -induced MHC-II expression in thyroid cells in vitro — representing a potential pharmacological strategy to reduce aberrant TFC antigen presentation [99].

5.3 Pan-MCF Therapeutic Strategies — Targeting Shared Upstream Nodes

The most transformative implication of the MCF framework is the identification of therapeutic targets that are shared across multiple diseases because they operate at common upstream nodes in the MCF cascade. We identify five pan-MCF therapeutic strategies of particular promise.

Vitamin D / VDR agonism: VDR signaling upregulates claudin-1, occludin, ZO-1, and filaggrin transcription across multiple epithelial tissues through VDRE-binding in the promoter regions of these genes [76,106]. Epidemiological evidence consistently associates vitamin D deficiency with elevated risk for MS, IBD, RA, psoriasis, and AITD, and interventional data (though heterogeneous in quality) suggest that vitamin D optimization reduces TJ protein loss in intestinal and epidermal epithelium [76,106]. Optimized vitamin D supplementation protocols (targeting 25-OH-D3 levels of 40–60 ng/mL) as a Stage 1 MCF preventive intervention in genetically at-risk individuals represent an immediately implementable, low-risk pan-MCF strategy.

Microbiome modulation: SCFA-producing commensals — particularly *Faecalibacterium prausnitzii*, *Akkermansia muciniphila*, and *Roseburia intestinalis* — stabilize intestinal TJ protein expression, maintain mucus layer integrity, and reduce systemic LPS translocation through the Toll-like receptor-MLCK-TJ axis [29,53,105]. Interventions targeting these microbial populations (dietary prebiotic optimization, targeted probiotic supplementation, FMT) act at Stage 1 of the MCF cascade in IBD, celiac, SS, MS, RA, and SLE simultaneously — reflecting the pan-autoimmune barrier-protective role of a healthy commensal microbiome.

Type I IFN pathway modulation: Anifrolumab (anti-IFNAR1) has been approved for SLE and is under investigation in SS — both diseases in which the type I IFN pathway is the primary Stage 3 amplifier of MCF-derived antigen spillover [113]. The possibility that anifrolumab or related IFNAR1 antagonists may have pan-MCF utility across psoriasis, MS, and RA (where type I IFN also plays significant Stage 3 roles) merits formal investigation in cross-disease trials. STING pathway inhibition (MSA-2, diABZI class inhibitors) is at an earlier development stage but represents a more specific molecular target at the innate sensing node [121].

JAK-STAT inhibition as barrier stabilizer: Beyond their established roles in suppressing pro-inflammatory cytokine signaling (Stages 3-5), JAK inhibitors — tofacitinib, upadacitinib, baricitinib, filgotinib — upregulate intestinal epithelial TJ protein expression through PTPN2-mediated JAK/STAT pathway normalization [70,102]. This dual mechanism — immune suppression plus barrier restoration — is mechanistically coherent with the MCF framework and likely contributes to the exceptional mucosal healing rates seen with JAK inhibitors in IBD beyond what cytokine suppression alone would predict [102,123]. Baricitinib's emerging efficacy across RA, AITD, alopecia areata, and IBD may partly reflect this pan-MCF TJ-stabilizing mechanism [17,102].

Omega-3 fatty acid membrane stabilization: EPA (eicosapentaenoic acid) and DHA (docosahexaenoic acid) are incorporated into cell membrane phospholipids, increasing membrane fluidity and reducing the production of inflammatory eicosanoids from arachidonic acid through competitive inhibition [124]. In the context of the MCF framework, omega-3 enrichment of acinar cell, intestinal epithelial, synovial, and brain endothelial membranes — which are the physical substrate of barrier construction — directly improves the material properties of barrier membranes that must withstand mechanical and osmotic stress without failure. Epidemiological associations

between omega-3 intake and reduced risk for RA, IBD, MS, and SLE [124], while not definitive, are consistent with a pan-MCF membrane-stabilizing mechanism.

5.4 Precision MCF Medicine: Barrier Biomarkers as Disease Stratification Tools

The MCF framework implies the existence of a measurable Stage 1 signature — a set of barrier integrity biomarkers that reflect the upstream MCF events before or alongside clinical autoimmunity — whose measurement could stratify patients for barrier-targeted therapy and provide earlier treatment windows than current symptom- or autoantibody-based approaches allow.

We propose the conceptual framework of a **Barrier Integrity Index (BII)** — a composite of clinically accessible barrier biomarkers weighted by disease-context relevance: (1) serum zonulin (reflecting gut TJ integrity — relevant across IBD, celiac, SLE, MS, RA, SS); (2) anti-AQP5 autoantibodies (salivary gland membrane integrity — SS-specific); (3) serum/CSF neurofilament light chain (NFL) and GFAP (neuronal and astrocytic barrier integrity — MS-specific); (4) plasma citrullinated histone H3 (NETosis-mediated nuclear membrane failure — RA/SLE-relevant); (5) plasma cell-free dsDNA (cfDNA) concentration and the interferon score (nuclear/mitochondrial membrane clearance failure — SLE-specific); and (6) transepidermal water loss (TEWL) measurement (stratum corneum barrier function — psoriasis/atopy-relevant) [47,56,71,72,95].

Significant standardization challenges remain for a functional BII: serum zonulin measurement is complicated by assay variability between commercial ELISA platforms that detect different fragments of the zonulin/haptoglobin-2 family; normal reference ranges for most barrier biomarkers in pre-disease populations are incompletely defined; and tissue-specific vs. systemic barrier readouts require development of separate reference ranges for different clinical contexts [71,72]. Nevertheless, the conceptual and translational

value of establishing barrier integrity measurement as a routine component of autoimmune disease assessment — parallel to the established role of autoantibody measurement at Stage 5 — is a logical and urgent priority for the field.

5.5 Combination Strategy: Barrier Restoration Plus Immune Modulation

Neither pure barrier restoration nor pure immunosuppression is sufficient as a monotherapeutic strategy for established autoimmune disease with a mature Stage 5 amplification loop. The barrier, damaged by autoantibodies and inflammatory cytokines, cannot be fully restored while inflammatory load remains high; and immunosuppression, while reducing the inflammatory insult to the barrier, does not repair the primary structural defects in TJ proteins, aquaporins, flaggrin, or nuclear membrane clearance mechanisms.

We propose a temporal three-phase combination strategy: **Induction (0-12 weeks)** — intensive immune-targeted therapy (biologics or JAK inhibitors) to reduce the inflammatory load that is actively damaging barrier structures through Stage 5 → Stage 1 feedback. This creates the permissive inflammatory environment in which barrier restoration is achievable. **Consolidation (3-12 months)** — barrier-restoration therapy deployed alongside de-escalating immune suppression: disease-specific interventions (larazotide, AQP gene therapy, PAD4 inhibitors, tapinarof, selenium) address the primary Stage 1 MCF defect, reducing antigen spillover and beginning to interrupt the autoantibody amplification cycle. **Maintenance (ongoing)** — microbiome optimization, vitamin D normalization, dietary antigen management, and omega-3 support to maintain barrier homeostasis and prevent recurrence of the environmental second hits that originally precipitated Stage 1 failure.

This approach mirrors the existing and validated therapeutic logic of celiac disease: gluten elimination (Stage 0 antigen blockade) combined with

nutritional restoration (addressing selenium/vitamin D/micronutrient deficiencies that compromise epithelial repair) achieves mucosal healing and, ultimately, serological remission — a trajectory not achievable by gluten elimination alone or by nutritional support without gluten restriction [89,119]. The MCF framework predicts that analogous temporal combination strategies should be equally logical and potentially as effective across the other seven diseases.

6. Discussion

The MCF framework presented in this paper constitutes a formal attempt to provide a unified mechanistic account of the shared pathological architecture underlying eight phenotypically diverse autoimmune diseases. In this Discussion, we synthesize the framework's explanatory power, address its principal limitations, and outline the research priorities its predictions generate.

Explanatory power — familial co-aggregation: The well-documented familial co-aggregation of autoimmune diseases — where first-degree relatives of patients with one condition carry elevated risk for multiple autoimmune phenotypes — is predicted by and coherent with the MCF framework. Families share not merely immune-regulatory alleles but barrier integrity gene variants (*FLG*, *PTPN22*, *NOD2*, *DNASE1L3*) and environmental exposures (microbiome composition, dietary antigens, infectious exposures) that determine Stage 1 vulnerability. The question is not which specific disease a genetically at-risk family member will develop but which of their barrier architectures will fail first — a question answered by tissue tropism (Section 4.5) rather than by separate disease mechanisms [12,15].

Explanatory power — sex bias: The female predominance of most autoimmune diseases (approximately 4:1 female-to-male ratio across the autoimmune spectrum, with some diseases at 9:1) has not been satisfactorily explained by immune-centric models. The MCF framework implicates estrogen as a modulator of multiple barrier systems: it upregulates claudin-5 in the BBB under homeostatic conditions but cyclically reduces intestinal claudin-1 during the luteal phase; it regulates AQP5 expression in exocrine glands; and it modulates NETosis propensity in neutrophils — creating sex-specific oscillations in barrier integrity that, across a lifetime of hormonal cycling, may create the cumulative barrier stress that exceeds tolerance thresholds in genetically susceptible women [107,108]. The abrupt onset or exacerbation of multiple autoimmune diseases during pregnancy, post-partum, and menopause — periods of maximal hormonal transition — is consistent with hormonally-driven barrier integrity perturbation as a disease-triggering event [107].

Explanatory power — comorbidity clustering: The MCF framework predicts that barrier failure in one tissue lowers the threshold for barrier failure in others, through two mechanisms. First, systemic type I IFN produced by the primary MCF site (e.g., salivary gland TLR7 activation in SS, or intestinal NLRP3 activation in IBD) circulates and destabilizes TJ proteins in remote tissues, creating multi-tissue barrier vulnerability [59,92]. Second, the inflammatory cytokine milieu of established autoimmunity (TNF- α , IL-6, IL-17) directly downregulates TJ and aquaporin expression in distal barrier tissues — predisposing established autoimmune patients to secondary barrier failures in new tissues, consistent with the observed pattern of autoimmune disease clustering within individuals [15,87,102].

Explanatory power — cross-class drug efficacy: The success of JAK inhibitors, anti-IL-17 antibodies, and anti-BAFF strategies across multiple distinct autoimmune diagnoses is coherently explained by the MCF

framework without requiring the separate ad hoc mechanistic explanations for each drug's cross-disease efficacy that organ-centric models necessitate. These drugs address shared nodes in the MCF cascade (Stage 3-5 amplification, Stage 5 → Stage 1 feedback loop) that are common across diseases by virtue of the shared upstream architecture — not because the diseases are less distinct than they appear, but because their divergent final stages share common biochemical effectors [16,17,102].

6.1 Limitations of the MCF Framework

Causality versus correlation: The most significant limitation of the MCF framework as currently presented is that, for several diseases, the temporal precedence of barrier failure over immune activation rests on correlational rather than definitively causal evidence. In SLE particularly, the question of whether nuclear envelope clearance failure drives IFN production or whether primary immune dysregulation drives nuclear membrane stress is not fully resolved by existing prospective cohort data, despite strong mechanistic inference from TREX1 and DNASE1L3 genetics [51,96,97]. The framework generates falsifiable predictions — specifically that longitudinal measurement of barrier biomarkers (BII) in genetically at-risk pre-clinical populations should consistently show Stage 1 marker elevation preceding Stage 5 autoantibody emergence by measurable intervals across all eight diseases — and we specifically invite prospective studies designed to test these predictions.

Reductionism risk: Assembling eight clinically and pathologically distinct diseases under a single mechanistic framework carries an inherent risk of oversimplification. We explicitly do not claim that MCF is the sole mechanism operating in any of these diseases, nor that disease-specific pathophysiological detail — the unique IL-35-producing plasmablast phenotype in SS, the NF-κB signaling hierarchy in IBD, the specific remyelination failure mechanisms in progressive MS, the platelet-driven

pathology in antiphospholipid syndrome complicating SLE — is irrelevant or reducible to MCF. We claim only that MCF constitutes the upstream vulnerability that shapes and channels disease-specific immune responses and tissue-specific destruction patterns — not that it is the sole or complete account of any disease's full pathological repertoire.

Therapeutic translation gap: While the MCF framework generates compelling mechanistic predictions for barrier-targeted therapy, clinical translation of Stage 1 interventions has been inconsistent. Larazotide acetate in celiac disease represents the clearest therapeutic success of barrier targeting [52]. AQP gene therapy in SS has not progressed beyond Phase 1 safety studies in radiation-damaged glands [112]. PAD4 inhibitors for RA have demonstrated pre-clinical efficacy but lack Phase 2 human validation [114]. This translational gap does not falsify the MCF model — therapeutic validation of mechanism is inherently challenging — but it does limit our current ability to assert that barrier-targeted intervention will prove clinically superior to immune-targeted intervention in diseases where downstream amplification loops are the dominant pathological driver at the time of clinical presentation.

Microbiome complexity: The role of the gut microbiome in MCF pathogenesis is bi-directional and incompletely characterized. The same microbiome that, in a healthy SCFA-producing state, maintains TJ integrity and reduces systemic antigen translocation, in a dysbiotic state becomes a primary driver of Stage 1 TJ failure across multiple diseases. The determinants of the microbiome-to-dysbiosis transition — antibiotic exposure, dietary composition, host genetics including *NOD2* variants, early-life microbial colonization patterns — add a dynamic and partially stochastic layer to the MCF framework that complicates linear mechanistic attribution [50,68,105].

6.2 Future Research Priorities

The MCF framework generates five high-priority research agendas whose execution would substantially advance the field:

First, **longitudinal pre-disease cohort studies** measuring BII biomarkers (serum zonulin, TEWL, cfDNA, NfL, citrullinated histones, anti-AQP5 antibodies) in first-degree relatives of autoimmune patients — stratified by HLA type — would directly test the temporal precedence hypothesis and establish Stage 1 → Stage 5 timelines for each disease [32,47].

Second, **human barrier challenge studies** (e.g., controlled gluten challenge in at-risk HLA-DQ2+ individuals with concurrent TJ monitoring; BMP6 serum level-correlated salivary AQP5 measurement in SS-risk individuals) would establish causal directionality with a level of evidence beyond the current observational data [52,89].

Third, **cross-disease clinical trials of pan-MCF agents** — specifically vitamin D optimization trials across multiple autoimmune diagnoses using barrier biomarker endpoints; microbiome restoration trials in pre-clinical autoimmune at-risk populations; and JAK inhibitor barrier-stabilization assessments using epithelial TJ protein biopsies as endpoints — would test therapeutic predictions of the framework [102,106].

Fourth, **BII development and validation studies** — establishing sensitivity, specificity, and minimal clinically important difference for each proposed BII component in well-characterized autoimmune cohorts — would enable clinical implementation of barrier-stratified therapeutic decision-making [72].

Fifth, **spatial transcriptomic and single-cell profiling at barrier interfaces** (enterocytes at the TJ, salivary acinar cells at the AQP5 membrane, brain microvascular endothelial cells at the BBB, TFCs at the follicular junction) in pre-disease, early-disease, and established-disease

states would provide the molecular resolution needed to characterize the MCF molecular signature at each disease stage with unprecedented precision [88,101].

7. Conclusion

We have presented the Membrane Construction Failure framework as a formal, mechanistically grounded account of shared upstream pathological architecture across eight major autoimmune diseases. The central proposition of the MCF framework is direct and falsifiable: *autoimmune diseases are barrier engineering disorders first and immune disorders second*. Their enormous clinical diversity — from dry mouth to demyelination, from inflammatory bowel to thyroid destruction — reflects the diversity of barrier architectures deployed across human tissues to maintain compartmental integrity, not fundamentally different disease mechanisms operating in parallel independence.

The five-stage MCF cascade — barrier construction or maintenance failure → antigen spillover → innate immune alarm activation → adaptive tolerance collapse → autoantibody/autoreactive T-cell amplification with barrier feedback damage — operates as a conserved pathological engine. Disease-specific tissue tropism is determined by the intersection of local barrier microenvironmental vulnerability, HLA-allele-restricted antigen presentation specificity, and tissue-restricted barrier gene expression — channeling the universal cascade into phenotypically distinct clinical presentations without requiring fundamentally distinct upstream mechanisms.

The shared genetic architecture of autoimmune diseases — clustering around HLA class II antigen presentation specificity, barrier integrity gene regulation (*PTPN22*, *VDR*, *NOD2*), and innate alarm threshold modulation

(*IRF5*, *STAT4*, *TREX1*) — is coherently interpreted within the MCF model as the heritable determinants of Stage 1 barrier vulnerability and Stage 3 innate amplification gain. Environmental modulators of MCF risk — microbiome dysbiosis, vitamin D deficiency, EBV infection, hormonal perturbation — all converge on the barrier layer before reaching the immune system, confirming the barrier as the primary vulnerable interface between environment and immune activation.

The therapeutic implications of the MCF framework are both specific and actionable. Barrier-targeted interventions — zonulin antagonists, AQP gene therapy, PAD4 inhibitors, selenium supplementation, epidermal barrier restoration, remyelination strategies, TREX1/efferocytosis augmentation — constitute an underexplored therapeutic class that addresses autoimmune disease at its mechanistic origin. These Stage 1 interventions are not alternatives to established immunotherapy; they are the missing component of a rational combination strategy that immunotherapy alone cannot provide, because immunosuppression does not rebuild barriers.

We call on the autoimmune research community to realign a meaningful proportion of research investment toward barrier integrity as a primary therapeutic and preventive target. The infrastructure required — barrier biomarker standardization, pre-clinical cohort establishment with longitudinal BII measurement, cross-disease barrier challenge models, and pan-MCF clinical trial consortia — is within current methodological reach. The potential reward is substantial: a unified prevention strategy for genetically at-risk individuals, targeting the Stage 1 MCF vulnerability before adaptive tolerance collapse and autoantibody amplification render the disease a self-perpetuating process requiring lifetime immunosuppression.

Autoimmune disease, reframed as a family of barrier engineering disorders, becomes both more conceptually coherent and more therapeutically

tractable. The membrane is where the story begins. It must be where the cure is sought.

Figure Legends

Figure 1. The MCF Five-Stage Cascade — Universal Architecture of Autoimmune Initiation. A schematic diagram depicting the five sequential stages of the Membrane Construction Failure cascade applicable across all eight diseases reviewed. Stage 1 (Barrier Failure) is shown at the apex of the cascade, with disease-specific molecular initiators (BMP6/AQP5, FLG deficiency, zonulin release, perforin poration, nuclear envelope clearance failure, aberrant MHC-II induction, cytokine-driven TJ dissolution) listed as parallel inputs. Stage 2 (Antigen Spillover) is illustrated with disease-specific antigens (SS-A/Ro, citrullinated peptides, LPS/flagellin, gliadin-tTG2 complexes, dsDNA/mtDNA, LL-37/self-DNA, myelin peptides, TPO/Tg) entering immune-surveilled space. Stage 3 (Innate Alarm) shows convergent innate sensors (TLR7/9, cGAS-STING, NLRP3, TLR4/5) with diverging outputs (type I IFN, NLRP3-IL-1 β /IL-18, complement MAC, NETosis). Stage 4 (Tolerance Collapse) depicts DC maturation, HLA-restricted T-cell activation, and Treg suppression. Stage 5 (Autoantibody Amplification) shows disease-specific autoantibodies feeding back to Stage 1 barrier via complement, Fc γ R activation, and direct barrier protein targeting. A bold red feedback arrow connects Stage 5 back to Stage 1, labeled "Self-Perpetuating MCF Loop." Color coding distinguishes Class A (epithelial, green), Class B (endovascular, blue), and Class C (intracellular membrane, orange) barrier types throughout.

Figure 2. Disease-Specific Barrier Architecture Map — Tissue Tropism Determinants. A schematic body diagram illustrating the eight barrier locations corresponding to the eight diseases reviewed. Each barrier site is

annotated with: (A) the primary barrier molecule (AQP5 for salivary glands in SS; claudin-5/occludin for BBB in MS; claudin-1/ZO-1 for gut in IBD/celiac; filaggrin/claudin-1 for skin in psoriasis; lamin network for nucleus in SLE; claudin-3/4 for thyroid follicle in AITD; synoviocyte/neutrophil plasma membrane for joint in RA); (B) the environmental second-hit associated with each site (iodine excess for thyroid; dysbiosis/LPS for gut; EBV/vitamin D deficiency for BBB; gliadin for small intestine; UV/FLG deficiency for skin; efferocytosis failure for nuclear envelope; BMP6 for salivary gland; perforin/PAD4 for synovium); and (C) the HLA allele directing disease-specific antigen presentation at each site. Inset panels show ultrastructural schematics of each barrier at the molecular level, highlighting the specific point of MCF (e.g., claudin strand dissolution, AQP5 mislocalization, MAC pore in myelin bilayer, ZO-1 phosphorylation and TJ gap formation).

Figure 3. Cross-Disease Molecular Convergence Heat Map. A color-coded heat map with the eight autoimmune diseases on the x-axis and 30 key MCF molecular players on the y-axis, illustrating the degree of involvement (absent, minor, moderate, primary) of each molecular actor in each disease. Rows include: tight junction proteins (claudin-1, claudin-4, claudin-5, occludin, ZO-1), aquaporin-5, PAD4/citrullination, NETosis/citrullinated histones, zonulin, filaggrin, lamin A/C, mtDNA spillover, cGAS-STING, TLR7, TLR9, TLR4/5, NLRP3, type I IFN (IFN- α/β), IL-17A, IL-23, TNF- α , complement MAC, NKG2D/IEL, HLA-DQ2/8, HLA-DR3/4, HLA-DR15, HLA-DRB1 shared epitope, PTPN22, IRF5, STAT4, IL23R, CTLA-4, and VDR. Columns (diseases) are clustered by MCF Stage 1 barrier class (Class A epithelial cluster: SS, IBD, celiac, psoriasis, AITD; Class B endovascular cluster: MS, RA; Class C intracellular cluster: SLE). The heat map visually demonstrates convergent molecular utilization across biologically distant diseases and highlights the type I IFN pathway and TJ proteins as pan-MCF convergence nodes.

Figure 4. MCF-Targeted Therapeutic Intervention Points — Disease-Specific and Pan-MCF Strategies. A therapeutic intervention map superimposed on the five-stage MCF cascade from Figure 1. Intervention arrows are color-coded by current development stage (approved: blue; Phase 2–3: green; Phase 1: orange; preclinical: red). Interventions are grouped by the MCF stage they target: Stage 0 (pre-cascade antigen detoxification — gluten protease therapy in celiac); Stage 1 — barrier restoration (larazotide acetate, AQP1 gene therapy, PAD4 inhibitors, tapinarof, selenium supplementation, remyelination strategies, FMT/microbiome restoration, vitamin D/VDR agonism, omega-3 membrane stabilization, TREX1 augmentation, DNase I replacement); Stage 3 — innate alarm suppression (cGAS-STING antagonists, NLRP3 inhibitors); Stage 4 — tolerance restoration (CTLA-4-Ig, tolerogenic DC strategies, antigen-specific tolerance protocols); Stage 5 — autoantibody reduction and loop interruption (anti-CD20, anti-BAFF, PAD4 inhibitors for NETosis). The Stage 5 → Stage 1 feedback arrow from Figure 1 is shown interrupted by Stage 1 barrier-restoration interventions, illustrating the proposed mechanism by which upstream targeting breaks the chronic self-perpetuating loop. A separate inset shows the proposed temporal combination strategy (Induction, Consolidation, Maintenance phases) with drug class assignments at each phase.

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